

MR Symbol Localization in Single Line Diagrams

Technical Product Requirements Document

Computer Vision Research + Engineering Specification

Architecture: Multi-Scale Chamfer Matching + PCA Subspace Verification

Methodology: Classical Computer Vision | One-Shot Template Matching

Constraint: Zero Training Data | Deterministic | CPU-Only

MR Symbol Localization in Single Line Diagrams - Technical PRD

Part 1: Executive Summary, Problem Definition, and Dataset Analysis

SECTION 1 EXECUTIVE SUMMARY

1.1 Problem Statement

This project addresses the localization of a specific Current Transformer (CT/MR) symbol across ten industrial Single Line Diagrams (SLDs) from the power systems engineering domain. Given exactly one query symbol image (MR_Symbol.png) and ten target SLD images, the system must automatically locate and return bounding boxes around every occurrence of the query symbol in every diagram.

1.2 Core Constraints

Constraint	Value
Query templates available	1
Target diagrams	10
Labeled training data	None
Annotated bounding boxes	None
Training epochs possible	N/A - no training set

1.3 Why Conventional Object Detection Is Not Ideal

Modern supervised object detectors (YOLO, Faster R-CNN, DETR, Mask R-CNN, Vision Transformers) require thousands of annotated examples to learn class-discriminative features. This project has:

- Zero annotations. No bounding box labels exist for any SLD.
- One template. A single reference image defines the target class.
- Ten images total. The entire corpus is ten diagrams - not ten thousand.
- No negative mining. Without labeled negatives, contrastive learning is infeasible.
- Determinism required. The system must produce identical results across runs - stochastic training is architecturally inappropriate.

Even few-shot detectors (e.g., Siamese networks, prototypical networks) require meta-learning episodes across many support/query pairs from multiple classes. With one class, one template, and zero annotations, these approaches collapse into template matching with unnecessary neural overhead.

1.4 Why This Project Is Fundamentally Different

This is not a conventional computer vision project. It is:

1. A symbol spotting problem - locating a known geometric pattern within structured technical drawings.
2. A one-shot localization problem - the entire "training set" is a single reference image.
3. A document image analysis problem - the inputs are engineering drawings with known conventions, not natural images.
4. A geometric matching problem - the symbol is defined by its shape, not its texture, color, or semantic context.

The dominant challenge is not image quality (the SLDs are clean, high-contrast, binary line drawings). The dominant challenge is discriminating the target symbol from visually similar neighboring symbols that share geometric sub-primitives (semicircular lobes, vertical stems, horizontal conductors).

SECTION 2 PROBLEM DEFINITION

2.1 Input Specification

Input	Description	Format
Query Symbol	`MR_Symbol.png` 161x103 pixels, RGBA, grayscale content on fully opaque white background	PNG
Target SLDs	`SLD1.png` through `SLD12.png` (10 files; SLD5, SLD6 absent)	PNG

2.2 Output Specification

For each SLD, produce a list of bounding boxes (x, y, w, h) where:

- (x, y) = top-left corner of the detected symbol occurrence
- (w, h) = width and height of the bounding box
- Each bounding box encloses one instance of the MR symbol

Additionally, produce annotated visualization images showing all detected bounding boxes overlaid on the original SLD.

2.3 Success Criteria

Metric	Target	Justification
Recall	≥ 0.90	Missing a real symbol is operationally dangerous
Precision	≥ 0.85	False positives waste engineering review time
F1-Score	≥ 0.87	Balanced performance requirement
IoU (per detection)	≥ 0.50	Standard PASCAL VOC localization threshold
Determinism	100%	Identical inputs must produce identical outputs
Explainability	Full pipeline transparency	Every detection must be traceable to a geometric score

2.4 Failure Criteria

- Recall < 0.70 on any individual SLD
- Precision < 0.50 (excessive false positives)
- System requires manual parameter tuning per SLD
- System requires GPU or deep learning inference
- Non-deterministic behavior across runs

2.5 Assumptions

1. All SLDs originate from the same engineering domain (power systems substation design).
2. The MR symbol represents a Current Transformer (CT) with a consistent geometric structure across all diagrams.
3. Symbol geometry is the primary discriminator - text labels are NOT part of the symbol definition.
4. SLDs are digitally generated (not scanned), resulting in clean line work.
5. The symbol appears connected to horizontal conductors (bus bars) in all diagrams.

2.6 Non-Assumptions (Requires Empirical Verification)

- Exact symbol count per SLD: Not assumed â?? must be determined empirically.
- Rotation range: Observed to be primarily 0° (horizontal orientation) but vertical orientation (90°) is present in SLD11. Full rotation range requires verification.
- Scale range: Symbol size relative to diagram varies. Exact scale factor distribution requires measurement.
- Symbol isolation: Symbols are NOT isolated connected components â?? they are topologically embedded in the bus structure (empirically verified in prior work).

2.7 Edge Cases

Edge Case	Description	Expected Handling
Overlapping text	Text labels ("MR", "2000/5", "Z2") directly adjacent to or overlapping the symbol	Text must not influence detection; purely geometric matching
Dense symbol chains	Multiple MR symbols in sequence with <1 symbol-width spacing	NMS must distinguish individual instances
Visually similar symbols	Symbols sharing the dual-semicircle lobe structure but with different stems/caps	PCA or structural verification must discriminate
Partial visibility	Symbols at diagram edges potentially cropped	Graceful degradation; report only fully visible instances
Scale variation	Different SLDs may render the symbol at different pixel sizes	Multi-scale matching required
Orientation variation	Most symbols are horizontal; SLD11 shows vertical arrangement	Multi-orientation matching required

3.1.1 Technical Properties

3.1.2 Geometric Structure (Observed from Image)

Horizontal conductor (top bar) Connects to bus; NOT part of symbol

Vertical stem + Cross/cap structure

Two adjacent downward semicircular lobes

1. Horizontal conductor (top): A horizontal line extending across the full width. This is the bus/conductor and is contextual â?? NOT part of the symbol's intrinsic geometry.
2. Vertical stem: A short vertical line descending from the conductor to the symbol body.
3. Cross/cap structure: A small cross or T-cap near the top of the stem (right side). This is a distinguishing feature.
4. Horizontal base bar: A thick horizontal line forming the base of the symbol body.
5. Two semicircular lobes: Two adjacent, downward-facing semicircles hanging from the base bar. These are the most visually distinctive feature.

3.1.3 Shape Characteristics

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Distinguishing features	(1) The dual-lobe structure, (2) the cross/cap, (3) the specific aspect ratio of lobes to stem
Potential confusion sources	Other symbols with semicircular elements, zigzag resistor symbols that share some edge structure, transformer symbols wi

3.1.4 Candidate Descriptors

Based on the observed geometry, the following descriptors are well-suited:

Descriptor	Suitability	Reason
Edge map (Canny)	High	Clean line drawing produces strong, unambiguous edges
Distance Transform	High	Enables efficient Chamfer matching against diagram edges
Shape Context	Medium	Captures spatial distribution of edge points, but sensitive to scale
Contour chain codes	Medium	The semicircular lobes produce distinctive contour sequences
HOG	Medium	Captures gradient orientation distribution of the lobe structure
Hu Moments	Low	Too global - cannot discriminate between symbols with similar mass distribution
SIFT/ORB keypoints	Low	Very few keypoints on simple line drawings

3.2 Forensic Analysis: Individual SLD Characteristics

3.2.1 Image Metadata Summary

SLD	Dimensions (W×H)	Aspect Ratio	File Size	Dark Pixel Ratio	Total Pixels
SLD1	1544 × 382	4.04	59,966 B	0.0292	589,808
SLD2	1024 × 678	1.51	92,423 B	0.0287	694,272
SLD3	1789 × 368	4.86	81,585 B	0.0354	658,352
SLD4	806 × 255	3.16	35,186 B	0.0402	205,530
SLD7	1778 × 819	2.17	169,448 B	0.0294	1,456,182
SLD8	901 × 537	1.68	58,386 B	0.0258	483,837
SLD9	1700 × 549	3.10	43,051 B	0.0135	933,300
SLD10	1581 × 915	1.73	136,435 B	0.0235	1,446,615
SLD11	769 × 698	1.10	56,541 B	0.0287	536,762
SLD12	871 × 112	7.78	11,503 B	0.0373	97,552

Key observations:

- All images are RGBA mode with grayscale content on white backgrounds.
- Dark pixel ratios are uniformly low (1.35%~4.02%), confirming sparse engineering drawings.
- Resolution varies significantly: from 97K pixels (SLD12) to 1.45M pixels (SLD7).
- Aspect ratios range from 1.10 (nearly square, SLD11) to 7.78 (extremely wide, SLD12).

3.2.2 Per-SLD Detailed Observations

SLD1 - 1544 × 382

Observed from image inspection:

- Layout: Single horizontal conductor line running left-to-right

- Complexity: Low â?? sparse layout with large whitespace
- Symbol density: Low â?? symbols are well-separated along the conductor
- MR symbol occurrences: Approximately 5 visible MR-labeled CT symbols along the main bus, plus 1 on a vertical branch below
- Text labels: "MR Y1 1200/5", "MR X1 1200/5", "MR X2 3000/5", "MR Y2 3000/5", "3000/5 X0 MR" â?? confirms multiple CT instances
- Visually similar symbols: Zigzag patterns (resistors/impedances), triangle symbols, surge arresters (SA)
- Noise: None â?? clean digital rendering
- Scan artifacts: None
- Skew/rotation: No skew. All MR symbols appear in standard horizontal orientation
- Potential FP sources: The zigzag transformer symbols share some edge structure with the MR lobes
- Potential FN sources: The vertically-oriented CT on the branch below may have different scale

Estimated MR count: ~6 (requires empirical verification)

SLD2 â?? 1024 Ã? 678

Observed from image inspection:

- Layout: Two horizontal bus bays (BAY#1 and BAY#2) with vertical interconnections
- Complexity: Medium-high â?? dense symbol population
- Symbol density: High â?? symbols tightly packed along both bus bays
- MR symbol occurrences: Numerous â?? each bay contains approximately 12 MR-labeled CTs. Labels visible: "MR 2000/5 Z2", "MR 2000/5 Y2", "MR 2000/5 X2", "MR 2000/5 X1", "MR 2000/5 Y1", "MR 2000/5 Z1" per section (712, 711, 722, 721 sections)
- Text labels: Dense text with colored labels (blue "MR" labels visible)
- Text density: High â?? labels for every symbol
- Visually similar symbols: Circuit breaker (G/B) boxes, motor (M) circles, disconnect switches, line traps
- Noise: None
- Skew: None. Includes rotated text labels (vertical text for cable specs)
- Engineering topology: Two parallel bus bays connected through breakers and VT sections
- Potential FP sources: G/B breaker rectangles, motor circles, disconnect switch elements
- Potential FN sources: Very dense packing may cause overlapping score basins in sliding window approaches

Estimated MR count: ~24 (requires empirical verification)

SLD3 â?? 1789 Ã? 368

Observed from image inspection:

- Layout: Single wide horizontal bus with sections 712, 711
- Complexity: Medium â?? this appears to be a zoomed/cropped view of one bay from SLD2
- Symbol density: Medium â?? symbols evenly distributed along the bus
- MR symbol occurrences: Approximately 12 MR-labeled CTs visible along the main conductor
- Text: "MR 2000/5" labels with position identifiers (Z2, Y2, X2, X1, Y1, Z1) for sections 712 and 711
- Visually similar symbols: G/B boxes, motor circles, VT section in center
- Noise: None
- Vertical structures: VT section (7T01VT) descends from center with branching topology
- Potential FP sources: The central VT section contains transformer symbols that may share edge structure with MR lobes
- Potential FN sources: Tightly packed CTs may merge in sliding window score maps

Estimated MR count: ~12 (requires empirical verification)

SLD4 â?? 806 Ã? 255

Observed from image inspection:

- Layout: Single horizontal conductor with section 722
- Complexity: Low â?? clean, isolated bus section
- Symbol density: Medium â?? symbols clearly separated
- MR symbol occurrences: ~6 visible MR-labeled CTs: "MR 2000/5 Z2", "Y2", "X2", "X1", "Y1", "Z1"
- Text: Clear labels, well-separated from symbols

- Visually similar symbols: G/B box, motor circles (M) at bus ends
- Noise: None
- Important: Clean validation target â?? all MR symbols clearly visible and well-separated
- Potential FP sources: Minimal â?? clean isolation
- Potential FN sources: Minimal â?? clear separation

Estimated MR count: ~6 (requires empirical verification)

SLD7 â?? 1778 Ã? 819

Observed from image inspection:

- Layout: Largest and most complex SLD â?? multi-section structure with 4+ bus sections (713, 712, 711, 723, 722, 721)
- Complexity: High â?? multiple bus bays, vertical interconnections, VT sections
- Symbol density: Very high â?? densely packed symbols across all sections
- MR symbol occurrences: Numerous â?? each section contains ~6 MR CTs, across 6 sections = ~36 total
- Text: Dense text labels throughout
- Engineering topology: Two major horizontal bus bars (7B02 and 7B01) connected via VT sections and breakers
- Visually similar symbols: G/B boxes, motor circles, disconnect switches, surge arresters, VT transformer symbols
- Noise: None
- Potential FP sources: High â?? complex topology creates many edge-dense regions
- Potential FN sources: Scale variation between sections possible

Estimated MR count: ~36 (requires empirical verification)

SLD8 â?? 901 Ã? 537

Observed from image inspection:

- Layout: Single bus section (713) with vertical VT branch
- Complexity: Medium â?? one bus section plus VT/SA branch
- Symbol density: Medium â?? symbols along one main conductor
- MR symbol occurrences: ~6 MR-labeled CTs on the main bus: "MR 2000/5 Z2", "Y2", "X2", "X1", "Y1", "Z1"
- Text: Clear labels
- Visually similar symbols: G/B box, motor circles, VT section with transformer, SA surge arrester
- Noise: None
- VT branch: Contains transformer symbols (zigzag), switch symbols â?? potential FP source
- Potential FP sources: Transformer/VT zigzag patterns
- Potential FN sources: Minimal â?? clean separation

Estimated MR count: ~6 (requires empirical verification)

SLD9 â?? 1700 Ã? 549

Observed from image inspection:

- Layout: Wide diagram with multiple vertical structures â?? breaker arrangements
- Complexity: Medium-high â?? different topology from other SLDs
- Symbol density: Medium â?? symbols distributed along a wider horizontal span
- MR symbol occurrences: Multiple CT-like symbols visible but with different geometric presentation â?? symbols appear smaller and the dual-lobe structure appears more as a downward "I?" shape
- Text: Minimal visible MR labels; section numbers 732, 731
- Visually similar symbols: Switch/breaker assemblies, motor circles, ground symbols, VT elements
- Noise: None
- Important: This SLD has a visually different rendering style â?? the CT symbols appear to have a different scale and aspect ratio relative to the bus structure
- Potential FP sources: Switch assemblies with curved elements
- Potential FN sources: Different scale may cause template mismatch

Estimated MR count: ~12â??18 (requires empirical verification â?? significant uncertainty due to different rendering style)

SLD10 â?? 1581 Ã? 915

Observed from image inspection:

- Layout: Two-bay structure similar to SLD2 â?? BAY#1 and BAY#2 with central VT section
- Complexity: Medium-high â?? structurally similar to SLD2
- Symbol density: High â?? dense symbol population on both bays
- MR symbol occurrences: Very similar to SLD2 â?? each bay contains ~12 MR CTs
- Text: Dense labels, blue-colored MR text
- Visually similar symbols: G/B boxes, motor circles, disconnect switches
- Noise: None
- Note: This appears to be a slightly different rendering of the same substation topology as SLD2

Estimated MR count: ~24 (requires empirical verification)

SLD11 â?? 769 Ã? 698

Observed from image inspection:

- Layout: VERTICAL arrangement â?? two parallel vertical buses (601, 602) with branching structure
- Complexity: Medium â?? fewer symbols but different orientation paradigm
- Symbol density: Medium â?? symbols distributed vertically
- MR symbol occurrences: ~4 visible MR-labeled CTs: "MR CT1 3000/5" and "MR CT2 3000/5" for both sections 601 and 602
- Text: Clear labels with "MR CT1" and "MR CT2" designations
- CRITICAL: The symbols in this SLD are oriented vertically â?? the horizontal base bar of the MR symbol is now vertical, and the lobes extend horizontally. This is a ~90Â° rotation from the standard orientation.
- Visually similar symbols: V/B boxes, motor circles (M), NC/NO switches, VT elements (601VT, 602VT)
- Noise: None
- Potential FP sources: V/B boxes share rectangular geometry
- Potential FN sources: 90Â° rotation means the template must be rotated to match

Estimated MR count: ~4 (requires empirical verification)

SLD12 â?? 871 Ã? 112

Observed from image inspection:

- Layout: Single narrow horizontal strip â?? one bus conductor
- Complexity: Very low â?? simplest SLD in the dataset
- Symbol density: Low-to-medium â?? symbols along a single line
- MR symbol occurrences: Multiple CT-like symbols visible â?? the dual-semicircle lobe pattern is clearly identifiable
- Text: Minimal â?? section label "733"
- Visually similar symbols: G/B box, motor circles, ground/switch elements
- Noise: None
- Important: Small image (871Ã?112 = 97K pixels) â?? symbols are proportionally small
- Potential FP sources: Minimal
- Potential FN sources: Small image size may cause scale issues

Estimated MR count: ~6 (requires empirical verification)

3.3 Dataset-Wide Analysis

3.3.1 Dataset Statistics

Statistic	Value
Total SLDs	10
Total pixel area	~7.1 million pixels
Mean dark pixel ratio	0.0282
Resolution range	97Kâ??1.46M pixels

Aspect ratio range	1.10~7.78
Image mode	All RGBA
Estimated total MR occurrences	~138 (significant uncertainty; requires ground truth)

3.3.2 Rotation Distribution

Orientation	SLDs	Percentage
0° (standard horizontal)	SLD1, SLD2, SLD3, SLD4, SLD7, SLD8, SLD9, SLD10, SLD12	90%
~90° (vertical)	SLD11	10%
Other angles	None observed	0%

Conclusion: The rotation distribution is bimodal ~ overwhelmingly 0° with one SLD showing 90° rotation. The system must handle at minimum 0° and 90° orientations. Continuous rotation is NOT observed and should not be the primary design target.

3.3.3 Scale Distribution

The MR symbol template is 161~103 pixels. The actual symbol size in diagrams varies:

- In dense SLDs (SLD2, SLD3, SLD7, SLD10): symbols appear approximately 25~40 pixels wide ~ roughly 4~6 smaller than the template.
- In sparse SLDs (SLD1, SLD4): symbols appear approximately 30~50 pixels wide ~ roughly 3~5 smaller than the template.
- In SLD9: symbols appear at a potentially different scale due to the different rendering style.
- In SLD12: symbols appear approximately 15~25 pixels wide due to the very compact image.

Conclusion: Scale variation is moderate (estimated 0.15~0.35 of template). Multi-scale matching is required with approximately 3~5 scale levels. Requires empirical calibration.

3.3.4 Symbol Occurrence Distribution

Estimated Count	SLDs
~4~6	SLD4, SLD8, SLD11, SLD12
~6	SLD1
~12	SLD3, SLD9
~24	SLD2, SLD10
~36	SLD7

Note: All occurrence counts are estimates from visual inspection. Ground truth annotation is required for precise evaluation.

3.3.5 Dataset Strengths

1. Extremely clean images ~ no noise, no scan artifacts, no compression artifacts. Dark pixel ratio < 5% across all SLDs.
2. High inter-document consistency ~ same engineering domain, same symbol family, same drawing conventions.
3. Binary-like contrast ~ near-perfect foreground/background separation simplifies binarization.
4. Structured layouts ~ symbols follow engineering grid patterns, reducing random spatial variation.
5. Consistent symbol identity ~ the MR symbol has a unique dual-semicircle lobe structure that is geometrically distinctive.

3.3.6 Dataset Weaknesses

1. No ground truth ~ zero labeled bounding boxes exist. Evaluation requires manual annotation.
2. Scale variation ~ the template is significantly larger than the actual symbols in diagrams.
3. Topological embedding ~ MR symbols are connected to bus conductors, preventing connected-component isolation (empirically proven in prior work).
4. Dense packing ~ in SLD2, SLD7, SLD10, symbols are very close together, creating overlapping score basins.
5. Orientation variation ~ SLD11 introduces 90° rotated symbols.

6. Rendering inconsistency â?? SLD9 appears to have a different rendering style/scale.

3.3.7 Dataset Risks

1. False positive hotspots: G/B breaker boxes, VT transformer zigzags, and surge arrester symbols share geometric sub-primitives with the MR symbol.
2. Score basin merging: In dense SLDs, adjacent MR symbols may produce merged score basins, causing under-counting.
3. Scale miscalibration: The 4â??6 scale difference between template and diagram symbols means scale estimation errors directly impact recall.
4. SLD9 anomaly: The visually different rendering style of SLD9 may require special handling or a wider scale search range.

3.3.8 Expected Matching Challenges

Challenge	Severity	Affected SLDs
Discriminating MR from G/B boxes	High	SLD2, SLD3, SLD4, SLD7, SLD8, SLD10
Discriminating MR from VT zigzags	Medium	SLD1, SLD3, SLD7, SLD8
Dense NMS in packed bus sections	High	SLD2, SLD7, SLD10
90° rotation handling	Medium	SLD11
Scale estimation for small symbols	Medium	SLD12
Different rendering style	Medium-High	SLD9

3.3.9 Expected Failure Modes

1. False positives on G/B breaker symbols: These rectangular boxes with internal structure share some edge density with MR symbols at certain scales.
2. Missed detections at extreme scales: If the scale search range is too narrow, symbols in SLD12 (very small) or SLD1 (potentially larger) may be missed.
3. Duplicate detections in dense sections: Adjacent MR symbols in SLD7 may not be properly separated by NMS.
4. Rotation failure on SLD11: If only 0° orientation is searched, all SLD11 symbols will be missed.
5. Template-to-diagram scale mismatch: The 161×103 template rendered at the wrong scale will produce poor Chamfer scores everywhere.

3.4 Mandatory Research Question: "What Problem Is This ACTUALLY?"

Analysis

Based on the forensic dataset inspection, this problem is:

Primarily: One-Shot Symbol Spotting via Geometric Template Matching

Evidence:

1. One-shot: Only one reference symbol is available. There is no training set, no validation set, no meta-learning support.
2. Symbol spotting: The task is to locate instances of a known geometric pattern in structured technical documents. This is the textbook definition of symbol spotting in Document Image Analysis (DIA).
3. Geometric template matching: The symbol is defined entirely by its geometric shape â?? straight lines and semicircular arcs. There is no texture, no color, no semantic context beyond geometry.

Secondarily: A Hybrid Problem combining:

- Multi-scale template matching (to handle the 4â??6 scale difference)
- Chamfer distance matching (to handle minor geometric deformations and anti-aliasing)
- Spatial non-maximum suppression (to consolidate dense overlapping detections)
- Semantic verification (PCA subspace or structural verification to discriminate from visually similar symbols)

NOT primarily:

- â?? Object detection (no training data)
- â?? Image retrieval (we know what we're looking for)

- â•? Graph matching (overkill for this geometric complexity)
- â•? Few-shot learning (requires meta-learning episodes we don't have)
- â•? Document understanding (we don't need to understand the diagram semantics)

This is a classical computer vision problem that should be solved with classical computer vision methods, augmented by principled verification stages to suppress structured false positives.

SECTION 4 LITERATURE AND METHOD SURVEY

This section analyzes every potentially relevant technique for MR symbol localization. Each technique is evaluated against the specific characteristics of this dataset: clean binary line drawings, one-shot template, scale mismatch, topologically embedded symbols, and a primary discrimination challenge against visually similar symbols.

4.1 Template Matching (Normalized Cross-Correlation)

How it works: Slides a template image across the target image, computing the normalized cross-correlation (NCC) at each position:

$$NCC(x,y) = \frac{\sum_{i,j} [T(i,j) - \bar{T}][I(x+i, y+j) - \bar{I}_{x,y}]}{\sqrt{\sum_{i,j} [T(i,j) - \bar{T}]^2 \cdot \sum_{i,j} [I(x+i, y+j) - \bar{I}_{x,y}]^2}}$$

Advantages: Simple, fast (FFT-accelerated), deterministic, well-understood.

Disadvantages: Fundamentally pixel-to-pixel assumes exact scale, rotation, and appearance match. No invariance to geometric deformation.

Expected performance on THIS dataset: Poor. The template is 161x103 pixels while diagram symbols are ~25x40 pixels. The scale mismatch makes direct NCC fail entirely. Even with rescaling, NCC is sensitive to the topological context (connected bus conductors alter the local pixel pattern).

Implementation complexity: Very low.

Suitability score: 2/10 Fundamental scale and context mismatch.

4.2 Multi-Scale Template Matching

How it works: Applies NCC at multiple scale levels by resizing the template to a pyramid of scales and running NCC at each level. Best match across scales is retained.

Advantages: Addresses scale variation. Still simple and deterministic.

Disadvantages: Computational cost scales linearly with number of scale levels. Still pixel-domain sensitive to anti-aliasing, line thickness variation, and topological context. No rotation handling without explicit rotation pyramid.

Expected performance on THIS dataset: Moderate. Would correctly localize symbols at the right scale level but will produce false positives on G/B boxes and VT elements that happen to have similar pixel density at certain scales. Cannot disambiguate geometric structure.

Implementation complexity: Low.

Suitability score: 4/10 Better than single-scale NCC but lacks geometric discrimination.

4.3 Chamfer Matching (Distance Transform Matching)

How it works: Operates in the edge domain. Computes the Distance Transform (DT) of the diagram's edge map, then overlays the template's edge pixels onto the DT. The Chamfer distance is the mean DT value sampled at template edge pixel locations:

$$D_{\text{chamfer}}(T, I) = \frac{1}{|E_T|} \sum_{p \in E_T} DT_I(p + t)$$

where E_T is the set of template edge pixels and t is the translation offset. Low Chamfer distance indicates good geometric

alignment.

Mathematical intuition: Rather than comparing raw pixels, Chamfer matching asks: "How far are the template's edges from the nearest diagram edges at this location?" This is fundamentally a geometric proximity measure rather than a pixel correlation measure.

Advantages:

- Edge-domain operation â?? invariant to fill patterns, textures, and background.
- Robust to minor deformation â?? the DT provides a smooth scoring landscape.
- Fast computation â?? DT is $O(n)$ via the Felzenszwalb algorithm; scoring is $O(|E_T|)$ per position.
- Naturally handles anti-aliasing â?? soft edge variations produce smooth DT gradients rather than hard mismatches.
- Proven in engineering drawing analysis â?? this is the classic approach for symbol spotting in CAD/technical drawings.

Disadvantages:

- No inherent scale invariance â?? requires multi-scale search.
- Sensitive to edge clutter â?? dense diagram regions with many edges produce false low-distance basins.
- Asymmetric â?? measures template-to-diagram distance but not diagram-to-template. A simple diagram edge can produce low scores if it happens to be near all template edges.

Expected performance on THIS dataset: Very good. Empirically validated in prior work on similar MR symbols: Chamfer matching successfully identified 3/5 MR symbols on a single SLD with a mean distance of 1.0 pixel and 87% edge coverage ratio. The edge-domain approach is perfectly suited to these clean binary line drawings.

Implementation complexity: Medium.

Suitability score: 9/10 â?? Primary candidate. Proven on this exact domain.

4.4 Bidirectional Chamfer Matching

How it works: Extends standard Chamfer by computing BOTH forward (templateâ??diagram) and reverse (diagramâ??template) distances:

$$D_{\text{bidir}} = \alpha \cdot D_{\text{forward}}(T \text{ to } I) + (1-\alpha) \cdot D_{\text{reverse}}(I \text{ to } T)$$

Advantages: Addresses the asymmetry problem. The reverse distance penalizes "empty" matches where a simple edge happens to align with the template but has no additional structure.

Disadvantages: Reverse distance computation requires extracting the local diagram edge patch at each candidate position, adding computational cost.

Expected performance on THIS dataset: Excellent. The bidirectional score would help discriminate MR symbols from isolated edges or simple geometric structures that produce low forward-only Chamfer scores.

Implementation complexity: Medium.

Suitability score: 8/10 â?? Strong enhancement to standard Chamfer.

4.5 Hausdorff Distance

How it works: Measures the maximum minimum distance between two point sets:

$$H(A,B) = \max(\max_{\{a \in A\}} \min_{\{b \in B\}} d(a,b), \max_{\{b \in B\}} \min_{\{a \in A\}} d(a,b))$$

The partial Hausdorff distance uses a percentile instead of max, making it robust to outliers.

Advantages: Theoretically elegant shape distance. Robust to partial occlusion when using the partial variant.

Disadvantages: Extremely sensitive to outlier edge pixels. Computationally expensive for dense point sets. The max operation

makes it unstable for practical use with noisy edges.

Expected performance on THIS dataset: Moderate. The clean edges reduce outlier sensitivity, but the connected bus conductor introduces many "foreign" edge points in the matching window that would inflate the Hausdorff distance even for true matches.

Implementation complexity: Medium.

Suitability score: 5/10 â?? Theoretically sound but practically inferior to Chamfer for this application.

4.6 Shape Context Descriptors

How it works: For each point on a contour, computes a log-polar histogram of the relative positions of all other contour points. Matching is performed by finding the minimum-cost correspondence between two sets of shape context descriptors using the Hungarian algorithm.

Advantages: Rich shape representation. Invariant to translation and scale (with normalization). Captures both local and global shape structure.

Disadvantages: Computationally expensive â?? $O(n^3)$ for the Hungarian algorithm on n contour points. Sensitive to the number and distribution of sample points. Requires clean, isolated contours â?? which are NOT available in this dataset because symbols are topologically embedded in the bus structure.

Expected performance on THIS dataset: Poor-to-moderate. The fundamental requirement for isolated contours is not met. Extracting contours from a bus-connected region would include bus conductor edges, corrupting the shape context descriptor. Could work on post-proposal candidate patches if the symbol can be approximately isolated.

Implementation complexity: High.

Suitability score: 4/10 â?? Contour isolation problem is a fundamental blocker for primary use.

4.7 Contour Matching (Fourier Descriptors)

How it works: Represents closed contours as sequences of complex numbers, computes the DFT, and uses the magnitudes of the low-frequency coefficients as rotation/scale-invariant shape descriptors.

Advantages: Compact representation. Inherent invariance to rotation, scale, and starting point.

Disadvantages: Requires closed, isolated contours. Sensitive to contour extraction quality. Cannot handle partial shapes or connected structures.

Expected performance on THIS dataset: Poor. Same contour isolation problem as Shape Context. The MR symbol's contour includes the bus conductor when embedded in the diagram.

Implementation complexity: Medium.

Suitability score: 3/10 â?? Contour isolation failure.

4.8 Connected Component Analysis

How it works: Labels connected regions in a binary image, then filters by geometric properties (area, aspect ratio, solidity, etc.).

Advantages: Very fast. Provides natural region proposals. Simple to implement.

Disadvantages: Fundamentally assumes symbols are isolated connected components. This was empirically tested and proven to fail in prior work on this exact domain â?? MR symbols are topologically connected to the bus conductor and do NOT emerge as

standalone components even after wire suppression and morphological repair.

Expected performance on THIS dataset: Failed â?? empirically verified. In Stage 2/2.5/2.75 of the prior pipeline, connected-component analysis was systematically tested with wire suppression, morphological closing experiments, and relaxed filtering thresholds. The MR symbol never emerged as an isolated connected component.

Implementation complexity: Very low.

Suitability score: 1/10 â?? Empirically invalidated.

4.9 Morphological Analysis (Erosion, Dilation, Opening, Closing)

How it works: Applies structuring element operations to binary images to modify topology â?? closing gaps, removing noise, separating touching structures.

Advantages: Can recover broken contours, separate weakly connected components. Fast.

Disadvantages: Structuring element choice is critical and problem-specific. Aggressive operations reconnect suppressed bus conductors, creating giant merged components.

Expected performance on THIS dataset: Useful as preprocessing, not as detection. Empirically validated in Stage 2.75: controlled morphological closing with small kernels (3Ã?3, 5Ã?3) successfully recovered fragmented MR coil geometry but did NOT enable connected-component isolation.

Implementation complexity: Low.

Suitability score: 5/10 as preprocessing support â?? 1/10 as primary detection.

4.10 Skeleton Matching

How it works: Computes the morphological skeleton (medial axis) of binary shapes, then compares skeletons using graph-based or structural matching.

Advantages: Topology-preserving dimensionality reduction. Captures the essential structure of shapes without fill information.

Disadvantages: Sensitive to noise and boundary irregularities. The skeleton of a bus-connected MR symbol includes the bus conductor skeleton, making isolated comparison impossible without prior segmentation.

Expected performance on THIS dataset: Moderate as a verification step. Could be useful for comparing the skeleton structure of candidate patches against the template skeleton, but cannot serve as the primary localization mechanism.

Implementation complexity: Medium.

Suitability score: 4/10 â?? Useful only for post-detection verification.

4.11 Graph-Based Matching (Graph Edit Distance)

How it works: Represents symbols as attributed graphs (nodes = junction points, edges = connecting strokes) and computes the graph edit distance (GED) â?? the minimum cost sequence of edit operations to transform one graph into another.

Advantages: Theoretically the most structurally precise matching method. Captures topology, not just geometry.

Disadvantages: NP-hard â?? GED computation is computationally intractable for large graphs. Requires prior graph extraction from pixel images, which is itself a hard problem. Overkill for geometrically simple symbols.

Expected performance on THIS dataset: Unnecessarily complex. The MR symbol has low structural complexity (a few lines and semicircles). Graph matching would work but provides no advantage over simpler geometric methods while incurring exponential computational cost.

Implementation complexity: Very high.

Suitability score: 2/10 â?? Overkill and computationally impractical.

4.12 PCA-Based Verification

How it works: Builds a PCA subspace from augmented versions of the template (rotations, scales, morphological variants). Projects candidate patches into this subspace and measures reconstruction error. Low reconstruction error indicates the candidate lies on the "symbol manifold."

Mathematical intuition: PCA learns the principal modes of appearance variation for the symbol class. True symbols project well onto these modes; non-symbols (G/B boxes, zigzags) project poorly, producing high reconstruction error.

Advantages: Provides semantic verification orthogonal to geometric matching. One-shot augmentation-driven training requires no external data. Computationally cheap at inference (matrix multiplication).

Disadvantages: Requires careful centroid alignment to prevent PCA from learning alignment variance rather than structural variance. The augmentation strategy must be carefully designed. Sensitive to PCA dimensionality.

Expected performance on THIS dataset: Empirically validated as a powerful disambiguation tool. In Stage 4 of prior work, PCA subspace refinement successfully discriminated true MR symbols from G/B false positives when combined with proper score calibration (exponential normalization, $\hat{I} \pm 2.0$, 0.7/0.3 fusion weights).

Implementation complexity: Medium.

Suitability score: 8/10 â?? Proven second-stage verification.

4.13 SIFT (Scale-Invariant Feature Transform)

How it works: Detects scale-space extrema in the Difference-of-Gaussians (DoG) pyramid, computes 128-dimensional gradient histogram descriptors at stable keypoint locations. Matching is performed via nearest-neighbor descriptor distance with Lowe's ratio test.

Advantages: Invariant to scale, rotation, and affine distortion. Highly discriminative descriptors.

Disadvantages: Designed for natural images with rich texture. Line drawings produce very few stable keypoints because DoG extrema require local gradient contrast that simple lines don't provide.

Expected performance on THIS dataset: Very poor. The MR symbol is composed of thin lines and arcs â?? the DoG pyramid produces almost no stable extrema. Empirically, line drawings typically yield 0â??5 keypoints per symbol, which is insufficient for reliable matching.

Implementation complexity: Low (OpenCV provides full implementation).

Suitability score: 1/10 â?? Fundamentally unsuited to line drawings.

4.14 SURF (Speeded-Up Robust Features)

How it works: Similar to SIFT but uses Hessian-based detector and integral image approximations for speed.

Advantages: Faster than SIFT. Same invariance properties.

Disadvantages: Same fundamental problem as SIFT â?? requires texture-rich images.

Expected performance on THIS dataset: Very poor. Same limitations as SIFT on line drawings.

Suitability score: 1/10.

4.15 ORB (Oriented FAST and Rotated BRIEF)

How it works: Uses FAST corner detector with Harris corner measure for keypoint detection, BRIEF binary descriptors with learned rotation compensation.

Advantages: Very fast (binary descriptors). Free from patent issues.

Disadvantages: FAST corners require local intensity contrast. Binary descriptors have limited discriminative power for subtle shape differences.

Expected performance on THIS dataset: Poor. Line drawings produce sparse corners primarily at junction points (intersections with bus conductors). These junctions are shared across many symbol types, providing no discrimination.

Suitability score: 2/10.

4.16 AKAZE (Accelerated-KAZE)

How it works: Uses nonlinear scale space (Perona-Malik diffusion) for keypoint detection, Modified-Local Difference Binary (M-LDB) descriptors.

Advantages: Nonlinear scale space preserves boundaries better than Gaussian blurring. Better suited to structured images than SIFT/SURF.

Disadvantages: Still assumes sufficient local texture/gradient contrast for keypoint detection.

Expected performance on THIS dataset: Moderate. AKAZE's nonlinear diffusion preserves edges better, potentially producing more stable keypoints on line drawings. However, the fundamental sparsity problem remains â?? there are very few distinctive keypoint locations in the MR symbol.

Suitability score: 3/10 â?? Slightly better than SIFT/ORB but still insufficient for primary matching.

4.17 HOG (Histogram of Oriented Gradients)

How it works: Divides the image into cells, computes gradient orientation histograms per cell, normalizes across blocks. The resulting descriptor captures local edge/gradient structure.

Advantages: Captures the distribution of edge orientations â?? directly relevant for the MR symbol's mix of horizontal lines, vertical lines, and curved arcs. Translation-invariant within cells.

Disadvantages: Fixed-size descriptor â?? requires candidate patches to be extracted first (not a localizer). Sensitive to cell/block size parameters. No inherent scale or rotation invariance.

Expected performance on THIS dataset: Moderate as a verification feature. HOG descriptors of the MR symbol's dual-semicircle lobes would have a distinctive gradient orientation distribution (strong vertical gradients from the lobes, strong horizontal from the base bar). Could serve as a secondary verification feature but cannot localize symbols.

Implementation complexity: Low.

Suitability score: 5/10 â?? Useful for post-detection verification, not localization.

4.18 Siamese Networks (One-Shot Learning)

How it works: Trains a twin neural network that maps pairs of images into an embedding space where similar images are close and dissimilar images are far apart. Uses contrastive loss or triplet loss.

Advantages: Explicitly designed for one-shot/few-shot scenarios. Learns discriminative features from pairs.

Disadvantages: Requires training data — specifically, positive and negative pairs. With one template and zero annotations, there are no positive pairs and no principled way to construct negative pairs. Transfer learning from natural image domains (ImageNet) is unlikely to help because line drawings have fundamentally different statistics.

Expected performance on THIS dataset: Poor without external data. A Siamese network cannot be meaningfully trained on a single template image. Even with augmentation, the network would learn template-specific artifacts rather than general symbol recognition. Could potentially work if pre-trained on a large corpus of engineering drawing symbols, but no such corpus is available.

Implementation complexity: High (requires deep learning framework, GPU, training loop).

Suitability score: 2/10 — Data requirements not met.

4.19 Few-Shot Learning (Prototypical Networks, Matching Networks)

How it works: Meta-learns an embedding space and a distance function across many episodes of support/query tasks.

Advantages: Designed for limited data scenarios.

Disadvantages: Meta-learning requires many classes and many episodes — not just few examples per class. With one class and one template, there is no meta-learning structure to exploit.

Expected performance on THIS dataset: Not applicable. The meta-learning paradigm requires a distribution of tasks, which does not exist here.

Suitability score: 1/10 — Paradigm mismatch.

4.20 YOLO / Faster R-CNN / DETR / Mask R-CNN

How it works: Supervised object detection architectures that learn to predict bounding boxes and class labels from thousands of annotated training images.

Analysis and rejection:

Requirement	Available
Training images needed	1,000–10,000+
Training images available	0 (template is not a training image)
Annotated bounding boxes needed	5,000–50,000+
Annotated bounding boxes available	0
Class diversity needed	Multiple classes for discriminative learning
Classes available	1 (MR symbol only)

Verdict: Rejected. These architectures are fundamentally unsuitable for this problem. No amount of augmentation on a single template can substitute for the thousands of annotated scene-level training examples these detectors require. Transfer learning from COCO/ImageNet would transfer natural image statistics that are irrelevant to engineering line drawings.

Suitability score: 0/10 — Hard reject due to data requirements.

4.21 Vision Transformers (ViT, DINOv2)

How it works: Self-supervised or supervised transformers that learn patch-level representations through attention mechanisms.

Analysis: DINOv2 self-supervised features could theoretically provide good representations without labeled data. However, the features are learned from natural images (ImageNet-scale), and engineering line drawings lie far outside this distribution. Fine-tuning requires labeled data we don't have.

Suitability score: 1/10 â?? Domain gap is too large for zero-shot transfer.

4.22 Classical Feature Matching (Template Histogram Comparison)

How it works: Compares histograms of image properties (intensity, gradient magnitude, gradient direction) between template and candidate regions.

Advantages: Fast, simple.

Disadvantages: Discards all spatial structure â?? two regions with the same histogram but different spatial arrangements would score identically.

Expected performance on THIS dataset: Poor. Many symbols share similar edge density and gradient distributions. Spatial structure is the key discriminator.

Suitability score: 2/10.

4.23 Hybrid Pipeline: Chamfer Localization + PCA Verification

How it works: Uses multi-scale Chamfer matching as the primary localization mechanism, followed by spatial NMS for consolidation, and PCA subspace refinement for semantic verification.

Advantages: Combines the strengths of geometric edge matching (localization) with appearance manifold verification (discrimination). Each stage addresses a specific failure mode:

- Chamfer â?? spatial localization
- Multi-scale search â?? scale invariance
- Multi-orientation search â?? rotation handling
- NMS â?? duplicate consolidation
- PCA â?? semantic disambiguation from structured false positives

Disadvantages: Multi-stage pipeline requires careful parameter calibration. Computational cost of dense sliding-window Chamfer scales with image size and number of scale/orientation levels.

Expected performance on THIS dataset: Excellent â?? empirically validated. This exact architecture was implemented and tested in prior work (Stages 1â??4.5), achieving:

- Correct localization of true MR symbols
- Successful suppression of G/B false positives via PCA refinement
- Stable NMS consolidation of overlapping score basins
- Recall improvement via multi-template ensemble search

Implementation complexity: Medium-high.

Suitability score: 10/10 â?? Proven architecture with empirical validation on this domain.

SECTION 5 ?? ARCHITECTURE TRADE STUDY

5.1 Evaluation Criteria

Each candidate architecture is evaluated against seven criteria derived from the dataset analysis:

Criterion	Weight	Rationale
One-shot capability	Critical	Only one template exists
Scale handling	High	4??6? scale mismatch between template and diagram symbols
Edge-domain suitability	High	Binary line drawings are most naturally represented as edge maps
Discrimination power	High	Must distinguish MR from G/B boxes, VT zigzags, and other visually similar symbols
Computational feasibility	Medium	Must run on CPU without GPU requirements
Determinism	Critical	Must produce identical results across runs
Empirical validation	Critical	Preference for methods empirically tested on this domain

5.2 Comparison Matrix

Architecture	One-Shot	Scale	Edge-Dom	Discrimination	CPU-Only	Deterministic	Empirical	**Overall**
NCC Template Matching	??	?	?	?	??	??	?	2/10
Multi-Scale NCC	??	??	?	?	??	??	?	4/10
Chamfer Matching	**??**	**??**	**??**	**??**	**??**	**??**	**??**	**9/10**
Bidirectional Chamfer	??	??*	??	???	??	??	?	8/10
Hausdorff Distance	??	??*	??	?? ?	??	??	?	5/10
Shape Context	??	??	??	??	?? ?	??	?	4/10
Contour Matching	??	??	??	?? ?	??	??	?	3/10
CC Analysis	??	??	?	?	??	??	?? Failed	1/10
Skeleton Matching	??	?? ?	??	?? ?	??	??	?	4/10
Graph Edit Distance	??	??	??	???	?? NP-hard	??	?	2/10
PCA Verification	**??**	**??**	**?? ?	**???	**??**	**??**	**??**	**8/10**
SIFT/SURF	??	??	?	?	??	??	?	1/10
ORB	??	?? ?	?	?	??	??	?	2/10
AKAZE	??	??	?? ?	?? ?	??	??	?	3/10
HOG	??	?	?? ?	?? ?	??	??	?	5/10
Siamese Networks	?? ?	??	?	?? ?	?	?	?	2/10
Few-Shot Learning	?	??	?	?? ?	?	?	?	1/10

YOLO/RCNN/DETR	â•?	â??	â•?	â??	â•?	â•?	â•?	0/10
Vision Transformers	â•?	â??	â•?	â? i•	â•?	â•?	â•?	1/10
Chamfer + PCA Hybrid	**â??**	**â??**	**â??**	**â??â??**	**â??**	**â??**	**â??**	**10/10**

â?? = requires multi-scale search wrapper, which is a standard extension.*

5.3 Detailed Rejection Justifications

5.3.1 YOLO / Faster R-CNN / DETR / Mask R-CNN â?? REJECTED

Rejection reason: Data requirements fundamentally unmet.

These supervised detectors require 1,000â??50,000+ annotated training images. This project has:

- 0 annotated bounding boxes
- 1 template image (not a training image)
- 10 target images (not a training set)

Even with aggressive augmentation of the single template, supervised detectors cannot learn:

- Background context statistics (what is NOT the symbol)
- Hard negative examples (G/B boxes that look similar but are not MR symbols)
- Scale/aspect ratio distributions from real scene-level examples

Data augmentation of a single isolated template produces artificial training examples that do not reflect the actual in-context appearance of the symbol embedded within bus structures.

Strength of rejection: Definitive. No mitigation possible without external annotated data.

5.3.2 Siamese / Few-Shot Networks â?? REJECTED

Rejection reason: Meta-learning paradigm mismatch.

Few-shot learning requires:

- A meta-training distribution of tasks (many classes, many episodes)
- Support/query pairs from multiple categories
- A learned metric/embedding space

This project has one class, one template, and zero labeled scene-level examples. There is no meta-learning structure to exploit. Pre-training on natural image pairs (Omniglot, miniImageNet) would not transfer to engineering line drawings.

Strength of rejection: Strong. Could be revisited if a large corpus of engineering drawing symbol annotations becomes available.

5.3.3 SIFT / SURF / ORB â?? REJECTED

Rejection reason: Feature detection failure on line drawings.

These feature detectors are designed for natural images with rich texture gradients. Engineering line drawings consist of thin lines and arcs that produce:

- Very few stable keypoints (0â??5 per symbol)
- High positional instability in the scale-space pyramid
- Non-distinctive descriptors at junction points (shared across symbol types)

Strength of rejection: Strong. Fundamental domain mismatch between textured natural images and binary line drawings.

5.3.4 Connected Component Analysis â?? REJECTED

Rejection reason: Empirically invalidated.

Tested in Stages 2, 2.5, and 2.75 of prior work. The MR symbol does NOT emerge as an isolated connected component

because:

- It is topologically connected to the bus conductor
- Wire suppression removes horizontal conductors but does not sever the vertical stem connection
- Morphological closing experiments recovered coil continuity but did not enable component isolation

Even with relaxed filtering thresholds (allowing 4× the template area), zero additional MR candidates were recovered.

Strength of rejection: Definitive. Empirically proven to fail on this exact dataset.

5.3.5 Graph Edit Distance ??? REJECTED

Rejection reason: Computational intractability and over-engineering.

GED is NP-hard. For practical graph sizes (10⁴–50 nodes), approximate algorithms exist but are still $O(n^4)$ or worse. The MR symbol has low structural complexity that does not justify graph-theoretic reasoning. Chamfer matching captures the necessary geometric discrimination at a fraction of the computational cost.

Strength of rejection: Strong. Correct in theory but impractical and unnecessary for this symbol complexity.

5.3.6 Shape Context ??? REJECTED as primary method

Rejection reason: Contour isolation requirement not met.

Shape Context requires sampling points from isolated contours. The MR symbol is topologically embedded in the bus structure ??? there is no clean, isolated contour to sample from. Attempting to extract a local contour from a candidate patch would include bus conductor edges, corrupting the descriptor.

Strength of rejection: Strong for primary localization. Could potentially be used as a verification step on extracted candidate patches with manual contour cleaning.

5.3.7 NCC Template Matching ??? REJECTED as primary method

Rejection reason: Pixel-domain sensitivity to context.

NCC compares raw pixel values, which are influenced by:

- The connected bus conductor passing through the matching window
- Adjacent text labels
- Neighboring symbols within the matching window
- Anti-aliasing variations

In engineering drawings, the local pixel context around a symbol is highly variable (different neighboring symbols, different text labels, different bus orientations), making pixel-domain correlation unreliable.

Strength of rejection: Moderate. Multi-scale NCC could serve as a crude initial filter but should not be the primary architecture.

Why Chamfer over NCC: NCC operates in the pixel domain, making it sensitive to contextual variations (bus conductors, adjacent text, neighboring symbols). Chamfer operates in the edge domain, where only the geometric structure matters.

Why Chamfer over Hausdorff: Chamfer uses mean distance (robust to outlier edges from bus conductors). Hausdorff uses max distance (sensitive to any single distant edge pixel).

Why multi-scale: The template is 161x103 pixels; diagram symbols are ~25x40 pixels. Without multi-scale search, the scale mismatch guarantees failure.

Why multi-orientation: SLD11 contains 90° rotated symbols. Without at least 0° and 90° orientation search, these symbols will be missed entirely.

Stage 3: Spatial Non-Maximum Suppression

Why it exists: Dense sliding-window Chamfer matching produces overlapping score basins where many neighboring windows around a true symbol all produce low Chamfer scores. Without NMS, the top-K results would be dominated by multiple windows from the same symbol.

Why greedy NMS over clustering: Greedy NMS (sort by score, suppress within radius) is simpler, faster, and more deterministic than clustering-based approaches (DBSCAN, mean-shift). The suppression radius is naturally set to the template size.

Stage 4: PCA Subspace Verification

Why it exists: Chamfer matching alone cannot discriminate the MR symbol from other symbols that share geometric sub-primitives. The G/B breaker box, for example, has sufficient edge density to produce competitive Chamfer scores. PCA provides an orthogonal semantic verification signal that captures appearance manifold membership.

Why PCA over a neural classifier: PCA requires no labeled data - it constructs a symbol manifold from augmented versions of the single template. A neural classifier would require labeled positive and negative examples.

Why PCA over HOG: HOG provides a single descriptor vector without a notion of "manifold membership." PCA provides reconstruction error, which naturally measures how well a candidate fits the learned appearance space.

Why the exponential normalization and 0.7/0.3 fusion: Empirically calibrated in Stage 4 of prior work across 35 experimental configurations. Exponential normalization ($\hat{\pm}=2.0$) with 0.7 Chamfer / 0.3 PCA weighting produced the optimal ranking separation between true MR symbols and structured false positives.

6.3 Why This Architecture Is Optimal for This Dataset

1. Data-driven: Every design decision emerged from analyzing the actual images, not from following trends.
 2. One-shot compatible: No training data required beyond the single template.
 3. Edge-domain native: Operates in the natural representation space of binary line drawings.
 4. Empirically validated: The core Chamfer + NMS + PCA pipeline has been implemented and tested on similar MR symbols in prior work.
 5. Deterministic: No stochastic components. Identical inputs always produce identical outputs.
 6. Explainable: Every detection is traceable to a Chamfer score, a coverage ratio, a PCA reconstruction error, and a fused score.
 7. CPU-efficient: No GPU required. Distance Transform and sliding-window operations are well-optimized in OpenCV.
 8. Modular: Each stage can be independently tuned, validated, and debugged.
-

Output: Bounding boxes + Annotated SLDs + Detection metadata

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Output:

- template_gray ??? grayscale image (uint8)
- template_binary ??? binarized template (0/255)
- template_edges ??? Canny edge map (0/255)
- preprocessing_stats.json ??? quantitative metadata

Algorithm:

1. Load RGBA image, convert to grayscale: $\text{gray} = 0.299R + 0.587G + 0.114B$
2. Apply median filter (kernel=3) for anti-aliasing smoothing
3. Adaptive thresholding (Sauvola recommended, block_size=21, k=0.2) to produce binary mask
4. Invert if necessary so foreground (symbol) = 255, background = 0
5. Canny edge detection (low=50, high=150) on the binary or grayscale image
6. Save statistics: dimensions, foreground pixel count, foreground ratio, edge pixel count

Complexity: $O(W \times H)$??? single-pass per operation

Failure Modes:

- Incorrect binarization threshold ??? fragmented symbol edges
- Over-smoothing ??? loss of fine detail in the cross/cap structure

Mitigation: Parameterize all constants. Use adaptive thresholding to handle the anti-aliased grayscale edges.

7.2.2 Diagram Preprocessing (Stage 1b)

Purpose: Convert each SLD into clean binary and edge representations.

Input: SLD{N}.png (variable size, RGBA)

Output:

- diagram_gray ??? grayscale
- diagram_binary ??? binarized diagram
- diagram_edges ??? Canny edge map

Algorithm: Same pipeline as template preprocessing, applied per-SLD.

Key parameter: The adaptive threshold parameters may need to be consistent across all SLDs to ensure uniform binarization behavior. Since all SLDs have similar dark pixel ratios (1.4%~4.0%), a single parameter set should generalize.

7.2.3 Wire Suppression (Stage 1c)

Purpose: Suppress long horizontal conductor lines to reduce edge clutter and improve Chamfer matching precision.

Input: diagram_binary

Output: diagram_binary_no_wire ??? binary image with horizontal conductors suppressed

Algorithm:

1. Create a horizontal structuring element: $\text{kernel} = \text{np.ones}((1, W_{\text{wire}}), \text{dtype}=\text{np.uint8})$ where W_{wire} ??? 50~100 pixels
2. Apply morphological opening: $\text{wire_mask} = \text{cv2.morphologyEx}(\text{binary}, \text{MORPH_OPEN}, \text{kernel})$
3. Subtract wire mask: $\text{no_wire} = \text{binary} - \text{wire_mask}$
4. Re-extract edges from no_wire: $\text{edges_no_wire} = \text{cv2.Canny}(\text{no_wire}, \text{low}, \text{high})$

Mathematical justification: Morphological opening with a wide horizontal kernel preserves only structures that are at least W_{wire} pixels wide horizontally ??? i.e., bus conductors. Subtracting this mask removes the conductor while preserving symbols, which are narrower than W_{wire} .

Complexity: $O(W \times H)$ per morphological operation

Failure Modes:

- W_{wire} too small ??? incomplete conductor suppression
- W_{wire} too large ??? vertical text or symbol fragments suppressed
- Wire suppression severs the MR symbol's connection point, fragmenting it

Mitigation: Tune W_{wire} per-SLD or use the diagram-wide line length distribution to set it adaptively. Wire suppression is applied to the diagram ONLY if the template is processed without it.

7.3 Stage 2: Scale and Orientation Pyramid

7.3.1 Template Scale Pyramid (Stage 2a)

Purpose: Generate multiple scaled versions of the template edge map to handle the scale mismatch.

Input: `template_edges` (161x103)

Output: List of (`scale_factor`, `scaled_edge_map`) tuples

Algorithm:

1. Define scale range: `scales = np.linspace(s_min, s_max, N_scales)`
 - Recommended: `s_min=0.15`, `s_max=0.40`, `N_scales=10`
 - This produces templates from ~24x15 to ~64x41 pixels
2. For each scale factor `s`:
 - Resize template: `scaled = cv2.resize(template_edges, (0,0), fx=s, fy=s, interpolation=INTER_AREA)`
 - Re-threshold to ensure binary: `_, scaled = cv2.threshold(scaled, 127, 255, THRESH_BINARY)`
 - Extract edge coordinates: `edge_coords = np.argwhere(scaled > 0)`

Mathematical justification: The scale range [0.15, 0.40] covers the observed size ratio between template and diagram symbols. Ten scale levels provide sufficient resolution to find a good match without excessive computational cost.

Complexity: $O(N_{scales} \times W_{template} \times H_{template})$

7.3.2 Orientation Variants (Stage 2b)

Purpose: Generate rotated versions of each scaled template to handle orientation variation.

Input: Each scaled template edge map

Output: For each scale, a set of orientation variants

Algorithm:

1. Define orientations: `[0°, 90°, 180°, 270°]`
 - 0°: Standard horizontal (covers majority of SLDs)
 - 90°: Vertical orientation (covers SLD11)
 - 180°: Inverted horizontal (may exist for some symbols)
 - 270°: Inverted vertical
2. For each orientation:
 - Rotate template: `cv2.getRotationMatrix2D(center, angle, 1.0) + cv2.warpAffine`
 - Re-extract edge coordinates

Total template variants: $N_{scales} \times N_{orientations} = 10 \times 4 = 40$

7.4 Stage 3: Dense Chamfer Localization

7.4.1 Distance Transform (Stage 3a)

Purpose: Precompute the Distance Transform of the diagram edge map, enabling $O(1)$ distance lookup per edge pixel.

Input: `diagram_edges_no_wire` (binary edge map)

Output: DT $\hat{=}$ float32 distance transform image

Algorithm:

1. Invert edges: $\text{inv_edges} = 255 - \text{diagram_edges_no_wire}$
2. Compute Distance Transform: $\text{DT} = \text{cv2.distanceTransform}(\text{inv_edges}, \text{cv2.DIST_L2}, 5)$

Mathematical definition: $\text{DT}(x,y) = \min_{(x',y') \in E} \|(x,y) - (x',y')\|$ where E is the set of diagram edge pixels.

Complexity: $O(W \times H)$ via the Felzenszwalb & Huttenlocher exact Euclidean DT algorithm

Key property: $\text{DT}(x,y) = 0$ at edge pixels and increases smoothly away from edges. This creates a smooth scoring landscape for Chamfer matching.

7.4.2 Dense Sliding Window Search (Stage 3b)

Purpose: Exhaustively evaluate the Chamfer distance at every valid translation position.

Input: DT (diagram distance transform), template edge coordinates for each (scale, orientation) variant

Output: $\text{score_map} \hat{=}$ 2D array of Chamfer scores per position; one per (scale, orientation)

Algorithm:

For each template variant $(s, \hat{I})$:

```
For each valid position (tx, ty):
    chamfer_score = 0
    for each template edge pixel (ex, ey):
        diagram_x = tx + ex
        diagram_y = ty + ey
        chamfer_score += DT[diagram_y, diagram_x]
    chamfer_score /= |E_template|    # Mean distance

    # Coverage ratio: fraction of template edges within distance  $\hat{I}$ 
    coverage = count(DT[ty+ey, tx+ex] <  $\hat{I}$  for (ex,ey) in E_template) / |E_template|

    score_map[ty, tx] = chamfer_score
    coverage_map[ty, tx] = coverage
```

Coverage ratio threshold: $\hat{I} = 2.0$ pixels $\hat{=}$ a template edge pixel is "covered" if the nearest diagram edge is within 2 pixels.

Optimization opportunities:

1. Vectorize using NumPy: Extract all DT values at template edge locations simultaneously
2. Step size > 1 : Reduce computation by stepping 2×3 pixels (at the cost of localization precision)
3. Early termination: Skip positions where the first few edge pixels already have high DT values
4. Multi-scale coarse-to-fine: Run coarse search at lower resolution, refine at full resolution

Complexity: $O(N_{\text{scales}} \times N_{\text{orientations}} \times W_{\text{diagram}} \times H_{\text{diagram}} \times |E_{\text{template}}|)$

For a typical SLD (1500 $\times$ 800) with 40 template variants and ~ 100 edge pixels per template:

- $\sim 1500 \times 800 \times 40 \times 100 = 4.8$ billion operations
- With step=2: ~ 1.2 billion operations
- Feasible on modern CPU in minutes; parallelizable across scales

7.4.3 Score Map Aggregation (Stage 3c)

Purpose: Combine score maps across all (scale, orientation) variants to produce a single best-match map.

Input: All score maps and coverage maps from Stage 3b

Output:

- $\text{best_score_map} \hat{=}$ minimum Chamfer score at each position across all variants
- $\text{best_scale_map} \hat{=}$ scale that produced the best score at each position
- $\text{best_orientation_map} \hat{=}$ orientation that produced the best score
- $\text{best_coverage_map} \hat{=}$ coverage ratio at the best-scoring variant

Algorithm: For each position, retain the variant with the minimum mean Chamfer distance, subject to coverage ratio threshold:

```
For each (x, y):
    best_score = inf
    for each variant v:
        if coverage_map_v[y, x] >= MIN_COVERAGE and score_map_v[y, x] < best_score:
            best_score = score_map_v[y, x]
            best_scale = v.scale
            best_orientation = v.orientation
```

MIN_COVERAGE threshold: 0.65 â?? reject positions where fewer than 65% of template edges are within 2 pixels of a diagram edge. This prevents spurious low-distance matches in regions with only a few coincidental edge alignments.

7.4.4 Coverage Ratio Filtering (Stage 3d)

Purpose: Eliminate candidate positions with insufficient geometric evidence.

Input: best_score_map, best_coverage_map

Output: Filtered candidate list sorted by Chamfer distance

Algorithm: Retain only positions where:

1. coverage_ratio >= 0.65
2. mean_chamfer_distance < MAX_DISTANCE (e.g., 3.0 pixels)

7.5 Stage 4: Spatial Non-Maximum Suppression

7.5.1 Local Minima Extraction (Stage 4a)

Purpose: Identify spatially distinct score basins â?? positions that are local minima in the Chamfer score map.

Input: best_score_map

Output: List of candidate positions that are local minima

Algorithm:

1. Apply minimum filter with window size W_nms (e.g., 15?15):
local_min = scipy.ndimage.minimum_filter(score_map, size=W_nms)
2. Local minima are positions where score_map[y,x] == local_min[y,x]
3. Filter by coverage threshold

7.5.2 Greedy NMS (Stage 4b)

Purpose: Suppress duplicate detections from overlapping score basins.

Input: List of local minima with scores and coverage ratios

Output: Final detection list with duplicates removed

Algorithm:

```

Sort candidates by (mean_distance ASC, coverage_ratio DESC)
detections = []
for candidate in sorted_candidates:
    is_suppressed = False
    for existing in detections:
        dist = euclidean(candidate.position, existing.position)
        if dist < SUPPRESSION_RADIUS:
            is_suppressed = True
            break
    if not is_suppressed:
        detections.append(candidate)

```

SUPPRESSION_RADIUS: Set to $\max(\text{template_width}, \text{template_height}) * \text{best_scale}$ approximately the size of the detected symbol. This ensures that two detections of the same symbol are merged, while adjacent symbols remain separate.

7.5.3 Ranked Detection List (Stage 4c)

Each detection contains:

- Position (x, y) — top-left corner of the bounding box
- Bounding box (x, y, w, h) — derived from template size at the best-matching scale
- Mean Chamfer distance
- Coverage ratio
- Best matching scale
- Best matching orientation
- Rank (1 = best match)

7.6 Stage 5: PCA Subspace Verification

7.6.1 Template Augmentation (Stage 5a)

Purpose: Generate a diverse set of augmented template views to construct a representative appearance manifold from a single reference.

Input: `template_binary` (161×103)

Output: List of ~300–400 augmented binary template patches, all normalized to `PCA_TEMPLATE_SIZE` (e.g., 64×64)

Augmentation strategy:

1. Scale variations: [0.90, 0.95, 1.00, 1.05, 1.10] — 5 levels
2. Rotation variations: [-5°, -3°, -1°, 0°, +1°, +3°, +5°] — 7 levels
3. Morphological variations: [erode_1, original, dilate_1] — 3 levels (simulates line thickness variation)
4. Flip variations: [original, horizontal_flip] — 2 levels
5. Total: $5 \times 7 \times 3 \times 2 = 210$ base variants
6. Each variant undergoes centroid alignment before inclusion

Centroid alignment procedure:


```
def center_foreground_centroid(binary_img, target_size=(64,64)):
    coords = np.argwhere(binary_img > 0)
    cy, cx = coords.mean(axis=0)
    # Translate so centroid is at image center
    ty = target_size[0]//2 - cy
    tx = target_size[1]//2 - cx
    M = np.float32([[1,0,tx],[0,1,ty]])
    aligned = cv2.warpAffine(binary_img, M, target_size)
    return aligned
```

Mathematical justification: Centroid alignment ensures PCA learns structural variability (scale, rotation, thickness) rather than alignment variability (translation offsets). This was identified as a critical robustness fix in prior work.

7.6.2 PCA Manifold Construction (Stage 5b)

Purpose: Build a low-dimensional appearance subspace from the augmented templates.

Input: List of augmented template patches (each flattened to a 1D vector of length $64 \times 64 = 4096$)

Output: Fitted PCA model with $N_{\text{components}}$ (e.g., 10-20)

Algorithm:

1. Stack all augmented patches into a data matrix X of shape $(N_{\text{augments}}, 4096)$
2. Center: $X_{\text{centered}} = X - X.\text{mean}(\text{axis}=0)$
3. Optionally whiten: normalize by singular values
4. Fit PCA: $\text{pca} = \text{sklearn.decomposition.PCA}(n_{\text{components}}=N_{\text{components}}, \text{whiten}=\text{True})$
5. The principal components define the "MR symbol manifold"

Explained variance diagnostic: Check that the first 10 components explain $\approx 60\%$ of variance. If explained variance is too low, the augmentation may be too aggressive; if too high, it may be insufficient.

7.6.3 Candidate Patch Extraction and Scoring (Stage 5c)

Purpose: Extract image patches at each NMS-surviving detection location and project them into the PCA subspace.

Input: Detection list from Stage 4, original diagram image

Output: PCA reconstruction error and similarity score per detection

Algorithm:

For each detection $(x, y, w, h, \text{scale}, \text{orientation})$:

1. Extract patch from diagram: $\text{patch} = \text{diagram_binary}[y:y+h, x:x+w]$
2. Resize to PCA_TEMPLATE_SIZE: $\text{patch_resized} = \text{cv2.resize}(\text{patch}, (64,64))$
3. Apply centroid alignment: $\text{patch_aligned} = \text{center_foreground_centroid}(\text{patch_resized})$
4. Flatten: $v = \text{patch_aligned.flatten().astype(float)}$
5. Project into PCA space: $z = \text{pca.transform}([v])[0]$
6. Reconstruct: $v_{\text{hat}} = \text{pca.inverse_transform}([z])[0]$
7. Reconstruction error: $\text{RE} = \|v - v_{\text{hat}}\|^2$
8. PCA similarity: $S_{\text{pca}} = \exp(-\text{RE} / \hat{\sigma}^2)$ where $\hat{\sigma}^2$ is a calibration constant

7.6.4 Score Fusion and Re-ranking (Stage 5d)

Purpose: Combine geometric Chamfer scores with appearance PCA scores for final ranking.

Input: Chamfer scores and PCA similarities per detection

Output: Fused scores and re-ranked detection list

Algorithm:

1. Normalize Chamfer distances using exponential mapping:
 $S_{\text{chamfer}} = \exp(-\hat{d} / \hat{\sigma}_{\text{distance}})$ where $\hat{d} = 2.0$ (empirically calibrated)

2. Compute fused score:

$$S_{\text{fused}} = w_c \cdot S_{\text{chamfer}} + w_p \cdot S_{\text{pca}}$$

where $w_c = 0.7$, $w_p = 0.3$ (empirically calibrated)

3. Re-rank detections by S_{fused} descending

4. Apply final threshold: retain detections with $S_{\text{fused}} > T_{\text{final}}$

Mathematical justification for exponential normalization: Linear Chamfer distances have a narrow dynamic range for well-matched candidates (e.g., 1.0 vs 1.5 vs 2.0). Exponential mapping amplifies these small differences, creating sufficient scoring gap for the PCA component to influence ranking.

7.7 Stage 6: Output Generation

7.7.1 Bounding Box Extraction

For each final detection, compute the bounding box:

```
bbox = (x, y, w_template * best_scale, h_template * best_scale)
```

Optionally add padding: $\text{bbox_padded} = (\text{x-pad}, \text{y-pad}, \text{w}+2*\text{pad}, \text{h}+2*\text{pad})$

7.7.2 Visualization Overlays

Generate annotated images:

1. Detection overlay: Original SLD with green bounding boxes around detections, ranked labels
2. Chamfer heatmap: Color-coded score map showing geometric alignment quality
3. Per-detection crops: Individual cropped patches around each detection for inspection
4. PCA comparison grid: Template vs. candidate vs. reconstruction for each detection

7.7.3 Detection Metadata (JSON)

```
{
  "sld_filename": "SLD1.png",
  "num_detections": 5,
  "detections": [
    {
      "rank": 1,
      "bbox": [331, 174, 83, 53],
      "chamfer_distance": 1.00,
      "coverage_ratio": 0.87,
      "pca_similarity": 0.07,
      "fused_score": 0.72,
      "best_scale": 0.25,
      "best_orientation": 0,
      "confidence": "high"
    }
  ],
  "pipeline_parameters": {
    "scale_range": [0.15, 0.40],
    "n_scales": 10,
    "orientations": [0, 90, 180, 270],
    "min_coverage": 0.65,
    "nms_radius": 40,
    "pca_components": 10,
    "fusion_weights": [0.7, 0.3]
  }
}
```

7.8 Multi-Template Ensemble Enhancement (Optional Stage)

Purpose

Enhance recall by generating a small ensemble of template variants (scale, erosion/dilation, small rotations) and running independent Chamfer searches with each. The ensemble's score maps are aggregated via MIN (best score across templates at each position).

Template Ensemble (7 variants)

1. Baseline (original edges)
2. 0.95 $\times$ scale
3. 1.05 $\times$ scale
4. Light erosion (simulates thinner lines)
5. Light dilation (simulates thicker lines)
6. -3 $^{\circ}$ rotation
7. +3 $^{\circ}$ rotation

Aggregation

$$\text{ensemble_score}[y,x] = \min(\text{score_v1}[y,x], \text{score_v2}[y,x], \dots, \text{score_v7}[y,x])$$

Expected Benefit

Recovers symbols that the baseline template fails to match due to minor geometric differences (line thickness, rendering artifacts, slight scale misalignment). Empirically validated to improve recall from 3/5 to 4 $\frac{2}{5}$ in prior work on a single SLD.

Computational Cost

7 $\times$ the cost of single-template search. Feasible for 10 SLDs on modern CPU (estimated total: 10 $\frac{1}{2}$ to 30 minutes).

SECTION 8 IMPLEMENTATION PLAN

8.1 Repository Structure

```

Symbol Segmentor/
â??
â??â??â?? Data/
â??   â??â??â?? Symbol/
â??   â??   â??â??â?? MR_Symbol.png
â??   â??â??â?? SLDs/
â??       â??â??â?? SLD1.png
â??       â??â??â?? SLD2.png
â??       â??â??â?? SLD3.png
â??       â??â??â?? SLD4.png
â??       â??â??â?? SLD7.png
â??       â??â??â?? SLD8.png
â??       â??â??â?? SLD9.png
â??       â??â??â?? SLD10.png
â??       â??â??â?? SLD11.png
â??       â??â??â?? SLD12.png
â??
â??â??â?? src/
â??   â??â??â?? __init__.py
â??   â??â??â?? config.py           # All pipeline parameters (single source of truth)
â??   â??â??â?? preprocessing.py    # Stage 1: binarization, edge extraction, wire suppression
â??   â??â??â?? scale_pyramid.py     # Stage 2: multi-scale and multi-orientation template generat
â??   â??â??â?? chamfer_matching.py  # Stage 3: distance transform, sliding window, score maps
â??   â??â??â?? nms.py               # Stage 4: local minima extraction, greedy NMS
â??   â??â??â?? pca_verification.py  # Stage 5: augmentation, PCA manifold, score fusion
â??   â??â??â?? ensemble.py         # Stage 5b (optional): multi-template ensemble
â??   â??â??â?? visualization.py    # All debug/output visualizations
â??   â??â??â?? utils.py            # I/O, logging, JSON serialization, image loading
â??
â??â??â?? evaluation/
â??   â??â??â?? __init__.py
â??   â??â??â?? ground_truth/       # Manual annotations (PASCAL VOC or COCO format)
â??   â??   â??â??â?? SLD1.xml
â??   â??   â??â??â?? SLD2.xml
â??   â??   â??â??â?? ...
â??   â??â??â?? metrics.py          # Precision, Recall, F1, IoU computation
â??   â??â??â?? evaluate.py         # Main evaluation script
â??   â??â??â?? ablation.py        # Ablation study runner
â??
â??â??â?? outputs/
â??   â??â??â?? template/          # Preprocessed template outputs
â??   â??â??â?? diagrams/          # Per-SLD detection outputs
â??   â??   â??â??â?? SLD1/
â??   â??   â??   â??â??â?? preprocessing/
â??   â??   â??   â??   â??â??â?? chamfer_heatmaps/
â??   â??   â??   â??   â??â??â?? nms_results/
â??   â??   â??   â??   â??â??â?? pca_analysis/
â??   â??   â??   â??   â??â??â?? detections/
â??   â??   â??   â??   â??â??â?? detection_results.json
â??   â??   â??   â??â??â?? ...
â??   â??â??â?? summary/           # Cross-SLD aggregate results
â??
â??â??â?? main.py                  # Main entry point â?? runs full pipeline on all SLDs
â??â??â?? requirements.txt        # Python dependencies
â??â??â?? config.yaml             # User-facing configuration file
â??â??â?? README.md               # Project documentation

```

8.2 Module Breakdown

8.2.1 `src/config.py` Configuration Constants

```
# === PREPROCESSING ===
MEDIAN_KERNEL_SIZE = 3
ADAPTIVE_BLOCK_SIZE = 21
ADAPTIVE_C = 10
CANNY_LOW = 50
CANNY_HIGH = 150
WIRE_KERNEL_WIDTH = 50

# === SCALE PYRAMID ===
SCALE_MIN = 0.15
SCALE_MAX = 0.40
N_SCALES = 10
ORIENTATIONS = [0, 90, 180, 270]

# === CHAMFER MATCHING ===
COVERAGE_THRESHOLD_TAU = 2.0          # pixels
MIN_COVERAGE_RATIO = 0.65
MAX_CHAMFER_DISTANCE = 3.0
SLIDING_WINDOW_STEP = 2              # pixels

# === NMS ===
NMS_WINDOW_SIZE = 15
NMS_SUPPRESSION_RADIUS_FACTOR = 1.0  # 1/2 template size

# === PCA VERIFICATION ===
PCA_TEMPLATE_SIZE = (64, 64)
PCA_N_COMPONENTS = 10
PCA_WHITEN = True
PCA_TAU = 2000.0
CHAMFER_ALPHA = 2.0
FUSION_WEIGHT_CHAMFER = 0.7
FUSION_WEIGHT_PCA = 0.3
FUSED_SCORE_THRESHOLD = 0.10

# === ENSEMBLE ===
ENSEMBLE_ENABLED = True
ENSEMBLE_SCALES = [0.95, 1.00, 1.05]
ENSEMBLE_ROTATIONS = [-3, 0, 3]
ENSEMBLE_MORPHOLOGIES = ['erode', 'original', 'dilate']
```

8.2.2 `src/preprocessing.py`

Functions:

Function	Input	Output	Purpose
`load_image(path)`	File path	RGBA ndarray	Safe image loading with validation
`to_grayscale(img)`	RGBA/RGB ndarray	Grayscale ndarray	Channel conversion
`denoise(gray)`	Grayscale	Denoted grayscale	Median filtering
`binarize(gray)`	Grayscale	Binary (0/255)	Adaptive thresholding
`extract_edges(binary)`	Binary	Edge map (0/255)	Canny edge detection
`suppress_wires(binary)`	Binary	Binary (wires removed)	Morphological wire suppression

<code>`preprocess_template(path)`</code>	File path	Dict of all outputs	Full template pipeline
<code>`preprocess_diagram(path)`</code>	File path	Dict of all outputs	Full diagram pipeline

8.2.3 ``src/scale_pyramid.py``

Functions:

Function	Input	Output	Purpose
<code>`build_scale_pyramid(edges, scales)`</code>	Edge map, scale list	List of (scale, scaled_edges)	Generate scale levels
<code>`build_orientation_variants(edges)`</code>	Edge map	List of (angle, rotated_edges)	Generate rotation variants
<code>`build_full_pyramid(edges)`</code>	Edge map	List of (scale, angle, edges, edge_coords)	Full scale&orientation pyramid
<code>`extract_edge_coordinates(edges)`</code>	Binary edge map	Nx2 ndarray of (row, col)	Edge pixel extraction

8.2.4 ``src/chamfer_matching.py``

Functions:

Function	Input	Output	Purpose
<code>`compute_distance_transform(edges)`</code>	Edge map	Float32 DT image	DT computation
<code>`chamfer_score_at_position(dt, edge_coords, tx, ty)`</code>	DT, coords, position	(mean_dist, coverage)	Single-position score
<code>`dense_chamfer_search(dt, edge_coords, step)`</code>	DT, coords, step size	score_map, coverage_map	Full sliding window
<code>`aggregate_multi_scale(score_maps)`</code>	List of score maps	best_score, best_scale, best_orient	Cross-variant min
<code>`filter_by_coverage(score_map, coverage_map)`</code>	Maps, threshold	Filtered candidate list	Coverage filtering

8.2.5 ``src/nms.py``

Functions:

Function	Input	Output	Purpose
<code>`extract_local_minima(score_map, window)`</code>	Score map, window size	Candidate list	Local minima detection
<code>`greedy_nms(candidates, radius)`</code>	Candidate list, radius	Filtered detections	Spatial suppression
<code>`compute_bounding_boxes(detections, template_size, scales)`</code>	Detections, size info	Bbox list	Bbox computation

8.2.6 ``src/pca_verification.py``

Functions:

Function	Input	Output	Purpose
----------	-------	--------	---------

<code>`center_foreground_centroid(binary, size)`</code>	Binary patch, target size	Aligned patch	Centroid normalization
<code>`generate_augmented_templates(template)`</code>	Template binary	List of aligned patches	Augmentation pipeline
<code>`build_pca_subspace(templates)`</code>	List of patches	Fitted PCA model	PCA fitting
<code>`extract_candidate_patches(diagram, detections)`</code>	Diagram, detections	List of patches	Patch extraction
<code>`compute_pca_scores(pca, patches)`</code>	PCA model, patches	List of (recon_error, similarity)	PCA scoring
<code>`fuse_and_rerank(detections)`</code>	Detections with both scores	Re-ranked detections	Score fusion

8.2.7 ``src/visualization.py``

Functions:

Function	Purpose
<code>`save_preprocessing_overview(output_dir, results)`</code>	Side-by-side preprocessing stages
<code>`save_distance_transform_vis(output_dir, dt)`</code>	Color-coded DT visualization
<code>`save_chamfer_heatmap(output_dir, score_map)`</code>	Chamfer score heatmap
<code>`save_detection_overlay(output_dir, image, detections)`</code>	Bounding boxes on original image
<code>`save_detection_crops(output_dir, image, detections)`</code>	Individual cropped patches
<code>`save_pca_grid(output_dir, templates)`</code>	Augmented template gallery
<code>`save_pca_reconstruction(output_dir, original, reconstructed)`</code>	PCA reconstruction comparison
<code>`save_nms_overlay(output_dir, image, before, after)`</code>	Pre/post NMS comparison

8.3 Data Flow

```

main.py
â??
â??â?? preprocess_template("Data/Symbol/MR_Symbol.png")
â??  â??â?? returns: template_results {gray, binary, edges, stats}
â??
â??â?? build_full_pyramid(template_results.edges)
â??  â??â?? returns: pyramid [(scale, angle, edges, coords), ...]
â??
â??â?? generate_augmented_templates(template_results.binary)
â??  â??â?? returns: augmented_patches [aligned_64x64, ...]
â??
â??â?? build_pca_subspace(augmented_patches)
â??  â??â?? returns: pca_model (sklearn PCA)
â??
â??â?? FOR EACH SLD in Data/SLDs/:
  â??
  â??â?? preprocess_diagram(sld_path)
  â??  â??â?? returns: diagram_results {gray, binary, edges, edges_no_wire, stats}
  â??
  â??â?? compute_distance_transform(diagram_results.edges_no_wire)
  â??  â??â?? returns: DT (float32 image)
  â??
  â??â?? FOR EACH (scale, angle, template_edges, template_coords) in pyramid:
    â??  â??â?? dense_chamfer_search(DT, template_coords, step=2)
    â??    â??â?? returns: score_map, coverage_map
    â??
    â??â?? aggregate_multi_scale(all_score_maps)
    â??  â??â?? returns: best_score_map, best_scale_map, best_orient_map
    â??
    â??â?? extract_local_minima(best_score_map, window=15)
    â??  â??â?? returns: raw_candidates
    â??
    â??â?? greedy_nms(raw_candidates, radius=template_size*scale)
    â??  â??â?? returns: nms_detections
    â??
    â??â?? extract_candidate_patches(diagram_binary, nms_detections)
    â??  â??â?? returns: candidate_patches
    â??
    â??â?? compute_pca_scores(pca_model, candidate_patches)
    â??  â??â?? returns: pca_scores
    â??
    â??â?? fuse_and_rerank(nms_detections, pca_scores)
    â??  â??â?? returns: final_detections
    â??
    â??â?? save_all_outputs(output_dir, final_detections, visualizations)

```

8.4 Configuration File (config.yaml)


```
data:
  symbol_path: "Data/Symbol/MR_Symbol.png"
  sld_directory: "Data/SLDs/"
  output_directory: "outputs/"

preprocessing:
  median_kernel: 3
  adaptive_blocksize: 21
  adaptive_c: 10
  canny_low: 50
  canny_high: 150
  wire_kernel_width: 50

matching:
  scale_min: 0.15
  scale_max: 0.40
  n_scales: 10
  orientations: [0, 90, 180, 270]
  coverage_tau: 2.0
  min_coverage: 0.65
  max_distance: 3.0
  step_size: 2

nms:
  window_size: 15
  suppression_radius_factor: 1.0

pca:
  template_size: [64, 64]
  n_components: 10
  whiten: true
  tau: 2000.0
  chamfer_alpha: 2.0
  weight_chamfer: 0.7
  weight_pca: 0.3
  score_threshold: 0.10

ensemble:
  enabled: true
  n_templates: 7

visualization:
  save_heatmaps: true
  save_crops: true
  save_overlays: true
  save_pca_diagnostics: true
```

8.5 Dependencies (`requirements.txt`)

```
opencv-python>=4.8.0
numpy>=1.24.0
scipy>=1.10.0
scikit-learn>=1.3.0
scikit-image>=0.21.0
matplotlib>=3.7.0
pyyaml>=6.0
```

SECTION 9 Evaluation Strategy

9.1 Ground Truth Creation Protocol

9.1.1 Annotation Format

Use PASCAL VOC XML format for bounding box annotations. Each SLD gets one XML file:

```
<annotation>
  <filename>SLD1.png</filename>
  <size>
    <width>1544</width>
    <height>382</height>
  </size>
  <object>
    <name>MR_Symbol</name>
    <bndbox>
      <xmin>325</xmin>
      <ymin>168</ymin>
      <xmax>370</xmax>
      <ymax>200</ymax>
    </bndbox>
  </object>
  <!-- ... more objects ... -->
</annotation>
```

9.1.2 Annotation Procedure

1. Annotator: Domain expert with power systems engineering knowledge
2. Tool: Labellmg or CVAT (open-source annotation tools)
3. Bounding box definition: Tight bounding box around the MR symbol body (semicircular lobes + base bar + stem + cap). Excludes connected bus conductor. Excludes text labels.
4. Labeling rule: Annotate EVERY visible MR/CT symbol, including partially visible ones at edges (marked with difficult=1)
5. Quality check: Two independent annotations, resolve disagreements by consensus

9.1.3 Expected Annotation Effort

Item	Estimate
SLDs to annotate	10
Estimated total symbols	~138
Time per symbol	~10 seconds
Total annotation time	~25 minutes
Quality review	~15 minutes
Total effort	**~40 minutes**

9.2 Metrics

9.2.1 Detection Metrics (per-SLD and aggregate)

Metric	Formula	Threshold
--------	---------	-----------

True Positive (TP)	Detection with IoU ≥ 0.50 against any GT box	$\text{IoU} \geq 0.50$
False Positive (FP)	Detection with no GT box having $\text{IoU} \geq 0.50$	$\hat{a}??$
False Negative (FN)	GT box with no detection having $\text{IoU} \geq 0.50$	$\hat{a}??$
Precision	$\text{TP} / (\text{TP} + \text{FP})$	$\hat{a}??$
Recall	$\text{TP} / (\text{TP} + \text{FN})$	$\hat{a}??$
F1-Score	$2 \hat{A} (P \hat{A} R) / (P + R)$	$\hat{a}??$

9.2.2 Localization Metrics

Metric	Formula	Description
IoU	$\text{area}(\text{det} \cap \text{gt}) / \text{area}(\text{det} \cup \text{gt})$	Per-detection localization quality
Mean IoU	average IoU across all TPs	Overall localization precision
Centroid Error	$\ \text{center}(\text{det}) - \text{center}(\text{gt})\ _2$	Pixel-level position accuracy
Mean Centroid Error	average centroid error across TPs	Overall positioning accuracy

9.2.3 Score Distribution Metrics

Metric	Description
Score gap	Difference between lowest TP score and highest FP score
Score separability	AUC of TP vs FP score distributions
Threshold stability	Range of thresholds that maintain $\text{F1} \geq 0.85$

9.3 False Positive Analysis

For every FP detection, record:

1. Position in the SLD
2. What structure was detected (G/B box, VT element, text, conductor junction, etc.)
3. Chamfer score $\hat{a}??$ how geometrically similar was it?
4. PCA score $\hat{a}??$ did PCA correctly assign low similarity?
5. Failure stage $\hat{a}??$ where in the pipeline should this FP have been rejected?

Classify FPs into categories:

- Geometric FP: Structurally similar symbol (G/B, VT) $\hat{a}??$ requires stronger PCA discrimination
- Clutter FP: Random edge alignment $\hat{a}??$ requires stricter coverage threshold
- Scale FP: Wrong-scale match $\hat{a}??$ requires tighter scale bounds
- NMS failure: Duplicate of a TP $\hat{a}??$ requires NMS radius tuning

9.4 False Negative Analysis

For every FN, record:

1. Position in the SLD
2. Why was it missed?
 - Scale out of search range
 - Orientation not searched
 - Chamfer score too high (poor edge alignment)
 - Coverage ratio too low
 - Suppressed by NMS (merged with a neighboring detection)
 - PCA rejected (reconstruction error too high)
3. Best-matching score at the GT location $\hat{a}??$ was there a local minimum, just not strong enough?

9.5 Ablation Studies

Experiment	Variable	Expected Insight
No wire suppression	Wire suppression ON vs OFF	Quantify impact of wire removal on Chamfer precision
Single scale vs multi-scale	1 scale vs 10 scales	Quantify recall gain from multi-scale search
Single orientation vs multi	0Å° only vs {0,90,180,270}	Quantify recall gain on SLD11
Chamfer only vs Chamfer+PCA	PCA OFF vs ON	Quantify FP reduction from PCA verification
Fusion weight sweep	w_c from 0.0 to 1.0	Find optimal Chamfer/PCA balance
Coverage threshold sweep	MIN_COVERAGE from 0.5 to 0.9	Precision/recall trade-off
NMS radius sweep	Radius from 0.5Å? to 2.0Å? template size	Under/over-suppression analysis
Ensemble vs single template	Ensemble OFF vs ON	Recall improvement from multi-template
PCA components sweep	N from 5 to 30	Reconstruction quality vs overfitting

9.6 Benchmark Comparisons

Baseline	Description	Purpose
Multi-scale NCC	OpenCV `matchTemplate` with TM_CCOEFF_NORMED	Pixel-domain baseline
Single-scale Chamfer	Chamfer at one fixed scale	Ablation of multi-scale
Chamfer without PCA	Full pipeline minus Stage 5	Ablation of PCA
Full pipeline	All stages enabled	Target performance
Human performance	Manual annotation agreement	Upper bound

9.7 Evaluation Script Output

The evaluation script (evaluation/evaluate.py) produces:

```
{
  "aggregate": {
    "total_gt": 138,
    "total_detections": 142,
    "true_positives": 130,
    "false_positives": 12,
    "false_negatives": 8,
    "precision": 0.915,
    "recall": 0.942,
    "f1_score": 0.929,
    "mean_iou": 0.72,
    "mean_centroid_error_px": 3.2
  },
  "per_sld": {
    "SLD1": {"gt": 6, "det": 6, "tp": 6, "fp": 0, "fn": 0, "p": 1.0, "r": 1.0},
    "SLD2": {"gt": 24, "det": 25, "tp": 23, "fp": 2, "fn": 1, "p": 0.92, "r": 0.96},
    ...
  },
  "false_positive_analysis": [...],
  "false_negative_analysis": [...]
}
```

Note: The aggregate metrics shown above are projected estimates, not measured values. Actual performance requires ground truth annotation and empirical evaluation. These projections are based on prior

SECTION 10 RISK ANALYSIS

10.1 Technical Risks

#	Risk	Probabilit	Impact	Mitigation
T1	**Chamfer score ambiguity** : G/B boxes or VT elements produce Chamfer scores indistinguishable from true MR symbols	Medium	High	PCA verification stage provides orthogonal discrimination signal. Empirically validated to suppress G/B FPs with exponen
T2	**Scale estimation failure** : The 4x6 scale ratio is incorrectly estimated, causing the scale pyramid to miss the tr	Low	Critical	Use a wide scale range (0.15x0.40) with 10 levels. Validate on SLD4 (cleanest SLD) first to calibrate the range
T3	**NMS over-suppression** : Adjacent MR symbols in dense bus sections (SLD7) are merged into a single detection	Medium	High	Set NMS suppression radius conservatively (1.0x template size at detected scale). Validate on SLD2/SLD7 which have the
T4	**Computational timeout** : Dense sliding window on large SLDs (SLD7: 1.5M pixels x 40 variants) exceeds acceptable runt	Medium	Medium	Use step=2, coarse-to-fine search, early termination. Parallelize across scales. Budget ~3 minutes per SLD
T5	**Wire suppression over-aggressiveness** : Horizontal morphological opening removes vertical symbol elements alongside bu	Low	Medium	Wire suppression uses a wide horizontal kernel (50+ pixels). MR symbols are much narrower than this threshold. Verify wi
T6	**Anti-aliasing artifacts** : Grayscale anti-aliasing on symbol edges produces weak/thick Canny edges that differ between	Low	Low	Median filtering smooths anti-aliasing. Adaptive thresholding handles gradual intensity transitions. Coverage ratio (I)=

10.2 Dataset Risks

#	Risk	Probabilit	Impact	Mitigation
D1	**SLD9 anomaly** : The visually different rendering style of SLD9 causes systematic scale or geometric mismatch	Medium-High	High	Extend scale range if needed. Run SLD9 as a separate diagnostic case. May require additional scale levels or template th
D2	**SLD11 rotation failure** : 90° rotated symbols are missed because the rotation search is insufficient	Low	High	Explicitly include 90° (and 270°) in the orientation search. SLD11 serves as the validation case for rotation handling

D3	**Ground truth creation bias** : Manual annotations are inconsistent or incomplete, undermining evaluation validity	Medium	High	Use two independent annotators. Define clear annotation rules (tight bbox, exclude text, exclude bus). Resolve disagreement
D4	**Unknown symbol count** : The estimated ~138 total occurrences may be significantly wrong (could be 80 or 200)	Medium	Medium	Ground truth annotation is the mitigation. Estimates are clearly labeled as such in this PRD
D5	**Missing SLDs** : SLD5 and SLD6 are absent from the dataset. These may contain symbol configurations not represented in	Low	Low	The 10 available SLDs cover diverse topologies (single bus, double bus, vertical, sparse, dense). Missing SLDs are unlikely

10.3 Algorithmic Risks

#	Risk	Probability	Impact	Mitigation
A1	**PCA manifold collapse** : Augmented templates are too similar, causing PCA to capture insufficient variance (explained	Low	Medium	Monitor explained variance diagnostic. If collapse occurs, increase augmentation diversity (add noise, wider rotation range)
A2	**Chamfer-PCA fusion instability** : The 0.7/0.3 fusion weights calibrated on prior work may not generalize to all 10 SLD	Medium	Medium	Run the fusion weight sweep ablation on all 10 SLDs. If weights need per-SDL tuning, this violates the "no manual tuning"
A3	**Coverage ratio threshold sensitivity** : MIN_COVERAGE=0.65 may be too strict (missing true symbols) or too lenient (admits	Medium	Medium	Run coverage threshold sweep ablation. Plot precision/recall curve as a function of coverage threshold
A4	**Multi-template ensemble introduces new FPs** : Ensemble variants (eroded/dilated templates) match non-MR symbols better	Low	Medium	Track per-template attribution: if a template variant produces new detections, verify they are TPs. Disable pathological

10.4 Scalability Risks

#	Risk	Probability	Impact	Mitigation
S1	**Larger SLD corpus** : Scaling from 10 to 100+ SLDs increases computational cost proportionally	High (if project succeeds)	Medium	Computational cost is linear in the number of SLDs. No architectural redesign needed. Parallelize across SLDs using multithreading

S2	**Higher resolution SLDs** : Future SLDs at 4000Å?2000+ pixels quadruple the per-SLD search cost	Medium	High	Implement coarse-to-fine search: run at 0.5Å? resolution first, refine around candidate regions at full resolution
S3	**Multiple symbol types** : Extension to detect 5â??10 different symbol types simultaneously	High (logical next step)	High	Current architecture handles one symbol type. Multi-symbol detection requires running the pipeline per-symbol or redesign

10.5 Maintenance Risks

#	Risk	Probabilit	Impact	Mitigation
M1	**Parameter drift** : New SLD batches from different drawing conventions require parameter re-tuning	Medium	Medium	Externalize all parameters to config.yaml. Document which parameters are drawing-convention-dependent vs universal
M2	**Dependency updates** : OpenCV, scikit-learn, or NumPy API changes break the pipeline	Low	Medium	Pin dependency versions in requirements.txt. Use only stable, well-documented APIs
M3	**Knowledge loss** : The pipeline's design rationale is lost when the original developer moves on	Medium	High	This PRD serves as the primary knowledge document. All design decisions are justified. Code must include inline comments

SECTION 11    FUTURE ROADMAP

Phase 1: MVP (Weeks 1  2)

Objective: End-to-end pipeline producing bounding boxes on all 10 SLDs.

Deliverables:

- Preprocessing module (template + diagram)
- Single-scale Chamfer matching at one estimated scale
- Basic NMS
- Detection overlays for visual inspection
- Per-SLD detection JSON files

Success criteria: Pipeline runs end-to-end without crashes. Visual inspection shows detections near true MR symbols on at least 5/10 SLDs.

Phase 2: Multi-Scale Matching (Weeks 3  4)

Objective: Scale-invariant detection across all SLDs.

Deliverables:

- Scale pyramid generation (10 levels)
- Multi-orientation search (0  , 90  , 180  , 270  )
- Scale-aware NMS with adaptive suppression radius
- Coverage ratio filtering
- Chamfer heatmap visualizations

Success criteria: Detections on 8/10 SLDs including SLD11 (rotated). Recall > 0.70.

Phase 3: PCA Verification (Weeks 5  6)

Objective: Suppress structured false positives using appearance manifold verification.

Deliverables:

- One-shot template augmentation pipeline
- PCA manifold construction with centroid alignment
- Score fusion with exponential normalization
- Re-ranking and final thresholding
- PCA diagnostics (variance spectrum, reconstruction examples)

Success criteria: Precision improves from ~0.70 to ~0.85+ while maintaining recall. G/B false positives suppressed.

Phase 4: Multi-Template Ensemble (Weeks 7  8)

Objective: Improve recall by expanding geometric coverage with template variants.

Deliverables:

- 7-template ensemble generation
- MIN-aggregated score maps
- Per-template attribution diagnostics
- New minima emergence tracking

Success criteria: Recall improves to > 0.85. New true detections emerge from template variants that the baseline misses.

Phase 5: Ground Truth and Rigorous Evaluation (Weeks 9-10)

Objective: Create ground truth annotations and perform rigorous quantitative evaluation.

Deliverables:

- PASCAL VOC annotations for all 10 SLDs
- Automated evaluation script (precision, recall, F1, IoU)
- False positive/negative analysis reports
- Ablation study results (wire suppression, scale, orientation, PCA, ensemble)
- Benchmark comparison table

Success criteria: F1 $\geq$ 0.87 with mean IoU $\geq$ 0.50. All failure modes documented.

Phase 6: Pseudo-Label Generation and Active Refinement (Weeks 11-14)

Objective: Use high-confidence detections as pseudo-labels to refine the pipeline.

Deliverables:

- Confidence calibration: classify detections as high/medium/low confidence
- Pseudo-label export for potential supervised fine-tuning
- Hard negative mining: systematically collect FP examples for analysis
- Adaptive threshold selection per-SLD based on score distribution

Success criteria: Self-consistent pipeline where high-confidence detections can bootstrap evaluation without manual annotation.

Phase 7: Deep Learning Transition (If Justified) (Months 4-6)

Objective: Evaluate whether deep learning can improve upon the classical pipeline.

Prerequisites: This phase should ONLY be pursued if:

1. Ground truth annotations exist for all 10 SLDs (~138+ labeled bounding boxes)
2. The classical pipeline has reached its performance ceiling (documented ablation studies)
3. Additional SLDs can be obtained to increase the training set

Potential approaches:

- Siamese matching network: Train on pseudo-labeled crops from the classical pipeline
- Fine-tuned object detector: Use the 10 annotated SLDs with aggressive augmentation for a small-data regime
YOLO/EfficientDet
- Self-supervised pretraining: Pre-train on engineering drawing patches, then fine-tune for symbol detection

Critical constraint: The deep learning system must demonstrably outperform the classical pipeline on the same evaluation metrics. If it does not, the classical pipeline remains the production system.

SECTION 12 Final Recommendation

12.1 Recommended Architecture

Multi-Scale Multi-Orientation Chamfer Matching with PCA Subspace Verification

A 6-stage classical computer vision pipeline:

Preprocessing Scale/Orientation Pyramid Dense Chamfer Localization
 Spatial NMS PCA Verification Output Generation

12.2 Why It Is Optimal

1. Data-appropriate: Designed for the actual constraints (1 template, 10 diagrams, 0 annotations).
2. Edge-domain native: Operates in the natural representation space of binary engineering drawings.
3. Empirically grounded: Core components (Chamfer matching, PCA verification, NMS) have been implemented and tested on similar MR symbols in prior work.
4. Deterministic: No stochastic components. Bit-for-bit reproducible.
5. Explainable: Every detection has a traceable geometric score, coverage ratio, PCA similarity, and fused confidence.
6. CPU-efficient: Runs on standard hardware without GPU requirements.
7. Incrementally improvable: Each stage can be independently tuned and verified.

12.3 Expected Strengths

Strength	Evidence
High recall on standard-orientation SLDs	Chamfer matching + multi-scale search covers the dominant 0° orientation across 9/10 SLDs
Effective FP suppression	PCA verification empirically suppresses G/B breaker false positives
Scale robustness	10-level scale pyramid covers the observed 3-6 scale range
Rotation handling	4-orientation search covers both horizontal and vertical symbol placements
Dense symbol discrimination	Coverage ratio filtering + NMS handles packed bus sections

12.4 Expected Weaknesses

Weakness	Severity	Mitigation Path
SLD9 rendering anomaly may require extra calibration	Medium	Extend scale range; add morphological template variants
Computational cost for large SLDs (~3 min/SLD)	Low	Coarse-to-fine search; parallelization
Parameter sensitivity (coverage threshold, NMS radius)	Medium	Ablation studies to find robust operating point
No automatic threshold selection	Medium	Score distribution analysis; Otsu-like auto-thresholding on fused scores

Requires manual ground truth for rigorous evaluation

Medium

One-time ~40 minute annotation effort

12.5 Expected Detection Performance

Disclaimer: The following are informed projections, NOT measured results. Actual performance requires ground truth annotation and empirical evaluation.

Metric	Projected	Basis
Precision	0.82 ~ 0.92	PCA verification suppresses most structured FPs; residual FPs from dense regions
Recall	0.85 ~ 0.95	Multi-scale + multi-orientation + ensemble covers most symbol configurations; potential misses on SLD9
F1-Score	0.84 ~ 0.93	Balanced by multi-stage pipeline
Mean IoU	0.60 ~ 0.75	Scale estimation accuracy limits IoU; bounding boxes may be slightly larger or smaller than GT
Per-SLD worst case	Recall ~ 0.70	Anticipated on SLD9 (rendering anomaly) or SLD7 (extreme density)

12.6 Research Value

This project demonstrates several principles of significant research value:

1. Classical methods remain competitive: For structured document analysis with clean inputs and low data availability, principled classical CV outperforms data-hungry deep learning.
2. One-shot symbol spotting: The entire pipeline bootstraps from a single reference image, proving that one-shot geometric matching is viable for industrial applications.
3. Hybrid architecture validation: The Chamfer (geometric) + PCA (appearance) combination demonstrates how orthogonal verification signals can be combined for robust detection.
4. Empirical methodology: The staged development approach (preprocessing ~ proposal ~ matching ~ verification ~ calibration ~ ensemble) provides a rigorous template for future CV research projects.
5. Anti-hallucination engineering: Every design decision is grounded in observed data characteristics, not assumed performance.

12.7 Industrial Value

Dimension	Value
Automation potential	Reduces manual symbol counting from hours to minutes
Accuracy	Projected to match or exceed human counting accuracy on clean SLDs
Consistency	Deterministic ~ eliminates inter-operator variability
Scalability	Linear cost in number of SLDs; parallelizable
Explainability	Every detection is auditable ~ critical for engineering compliance
Deployment simplicity	Python-only, CPU-only, no GPU infrastructure required
Extensibility	Architecture supports additional symbol types via per-symbol templates
Integration readiness	JSON output format compatible with downstream engineering databases

12.8 Closing Statement

This PRD was created after a comprehensive forensic analysis of every image in the dataset. Every architectural decision â?? from the selection of Chamfer matching over NCC, to the rejection of YOLO and Siamese networks, to the specific PCA fusion calibration â?? is grounded in observed dataset characteristics and, where available, empirical validation from prior work on the same engineering domain.

The recommended pipeline is not the most modern approach. It is not the most exciting approach. It is the correct approach for this specific problem, this specific dataset, and these specific constraints.

The dominant challenge is not image quality â?? these are clean, high-contrast engineering drawings. The dominant challenge is geometric discrimination â?? distinguishing the MR symbol from other symbols that share geometric sub-primitives. The recommended architecture directly addresses this challenge through a principled hierarchy of geometric matching (Chamfer), spatial consolidation (NMS), and semantic verification (PCA).

Implementation can begin immediately using this document as the architectural specification.